Evidence-Validated Strategy Adaptation for Memory-Augmented Agents Under Distribution Shift

A Statistical Framework for Reliable Experience Validation, Applicability-Gated Retrieval, and Selective Memory Invalidation

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Abstract

Memory-augmented autonomous agents store past trajectories, verbal reflections, or induced workflows to improve sequential decision-making without parameter updates. However, most non-parametric agent memory architectures rely on single-episode admission and surface-similarity retrieval without statistical verification, explicit applicability preconditions, or distribution-shift invalidation. Under non-stationary task streams and noisy evaluator feedback, unvalidated experiential stores accumulate spurious coincidences and obsolete routines, causing severe negative transfer. We present **Evidence-Validated Strategy Adaptation (EVSA)**, a closed-loop statistical framework that models reusable agent strategies as explicit hypotheses governed by three epistemic tiers (`CANDIDATE`, `VALIDATED`, and `QUARANTINED`). EVSA combines (i) a discounted log-odds sequential validation heuristic motivated by Wald's Sequential Probability Ratio Test (SPRT) with discounted Bayesian Beta(α, β) lower credible bounds, (ii) multiplicative applicability-gated retrieval with re-verified Least General Generalization (Safe LGG) over task preconditions, and (iii) per-domain residual tabular CUSUM monitoring for evidence-based selective post-shift quarantine.

We evaluate EVSA in **ShiftBench-Sim v1.0**, a controlled, low-dimensional synthetic symbolic-procedural simulation (4-step discrete operator workflows, observable 5-bit binary state descriptors, 20-dimensional synthetic embeddings, and a heuristic non-LLM `CalibratedSymbolicLLMBackbone` isolated behind a capability-restricted `AgentEnvironmentInterface`; $>$ 360,000 simulated episodes across $N = 20$ primary seeds and $N = 30$ stability seeds) against four budget-matched baselines (*Static*, *Conventional Memory*, *Reflection*, and *Experience Replay*). On held-out unseen tasks—where frozen evaluation executes strictly `VALIDATED` strategies—EVSA achieves **54.50% ± 5.04%** success (55.17% ± 5.62% across $N = 30$ seeds), outperforming all baselines (25.65%–29.62%, paired Cohen's $d_z \in [4.31, 6.35]$, Holm-Bonferroni $p \leq 8.15 \times 10^{-14}$), including zero-shot compositional tasks (31.13% vs. 12.12%–17.94%, $p_{\text{Holm}} \leq 1.07 \times 10^{-6}$). A controlled execution-channel parity study confirms that +22.04 percentage points ($\sim 82\%$) of EVSA's +26.92 pp advantage over Experience Replay persists under identical stochastic prompt execution (49.62% vs. 27.58%, $p_{\text{Holm}} = 1.57 \times 10^{-13}$). Across 320-episode continual streams, EVSA reaches **65.21% ± 8.52%** while compacting memory to 21.50 validated strategies (13.14 KB). Following abrupt multi-domain concept shifts, EVSA achieves +20.80 **pp** post-shock adaptation gain while preserving **82.21%** accuracy with no measurable cross-domain degradation on unshifted control domains. Under symmetric feedback corruption ($\eta \in \{0.10, 0.25, 0.35\}$), EVSA limits rollout-level false acceptance to **1.34%–2.53%** (candidate-level false promotion 8.08%–11.32%), significantly reduces active memory contamination across all noise levels under pre-defined per-family Holm adjustment ($p_{\text{Holm}} \leq 0.0063$; 60/74 study-wide tests significant under simultaneous global 74-test pooling), and recovers to **74.90%–84.10%** once clean feedback resumes, while candidly exposing sample-complexity versus epistemic-purity trade-offs under high noise (65.44% FRR at $\eta = 0.35$) and short per-archetype horizons.

Keywords: Memory-augmented agents; Sequential hypothesis testing; Bayesian credible bounds; Applicability-gated retrieval; Concept drift detection; Selective memory invalidation; Continual adaptation.

1 Introduction

Autonomous reasoning agents increasingly augment fixed decision backbones with external episodic or procedural memory repositories to accumulate operational competence at test time [1–5]. Rather than updating model weights via gradient descent—which is often computationally prohibitive, API-restricted, or prone to catastrophic forgetting [6]—non-parametric memory agents record past interaction traces, self-critiques, or extracted action workflows and retrieve them via vector similarity when tackling related tasks [7–9].

Despite strong gains in stationary, clean-feedback environments, prevailing agentic memory designs rest on three fragile implicit assumptions:

1. **Single-Episode Epistemic Trust:** If an action trajectory or verbal reflection coincides with task completion once, it is immediately admitted as reusable knowledge without testing whether the outcome arose from genuine task-operator alignment, a lucky stochastic transition, or a noisy evaluator signal [10, 11].
2. **Semantic-Applicability Equivalence:** High cosine similarity between surface task descriptions is treated as sufficient evidence that two tasks share identical operational constraints, ignoring latent structural preconditions that invert optimal tool sequences [12, 13].
3. **Temporal Stationarity:** Stored experiences remain active indefinitely (or decay only by passive recency heuristics), lacking online statistical change-point detection to quarantine invalidated strategies when external APIs, policies, or environment dynamics shift [14].

When these assumptions fail simultaneously, memory transitions from an asset into a liability. Surface-retrieved stale workflows crowd out valid exploration, noisy feedback promotes suboptimal routines that compound downstream errors [11], and indiscriminate memory resets sacrifice valid competence on unaffected task domains.

To address these coupled failure modes in a principled, auditable setting, this paper introduces **Evidence-Validated Strategy Adaptation (EVSA)**. EVSA treats every induced workflow strategy $h = (\phi_h, \pi_h)$ not as a raw text trace, but as a falsifiable statistical hypothesis pairing an explicit conjunctive precondition ϕ_h with an executable action plan π_h. Strategies enter a probationary `CANDIDATE` tier, undergo multi-trial verification governed by a discounted log-odds sequential heuristic and Bayesian Beta(α, β) lower credible bounds, generalize conservatively across nuisance features via re-verified Least General Generalization (Safe LGG) [15], and remain continuously audited in the `VALIDATED` tier by per-domain residual tabular CUSUM drift detectors [16].

Because isolating memory-validation dynamics from live API non-determinism and pretraining contamination requires exact ground-truth control over latent preconditions, changepoints, and feedback corruption, we conduct our empirical investigation in **ShiftBench-Sim v1.0**, a controlled, low-dimensional synthetic symbolic-procedural simulation (4-step discrete operator tuples, observable 5-bit binary state vectors, 20-dimensional synthetic embeddings, and a heuristic `CalibratedSymbolicLLMBackbone`). We explicitly avoid claiming evaluation on live neural Large Language Models (LLMs) or unstructured natural-language precondition extraction; instead, ShiftBench-Sim v1.0 provides a capability-isolated algorithmic testbed where every statistical mechanism, failure boundary, and baseline asymmetry can be audited rigorously.

Summary of Verified Contributions.

- **Closed-Loop Epistemic Strategy Architecture:** We formalize EVSA across ten modular components uniting sequential hypothesis gating, Bayesian lower credible reliability estimation, precondition-gated retrieval with Safe LGG re-verification, and domain-scoped residual CUSUM invalidation.
- **Mathematically Audited Sequential & Capability-Isolated Formulations:** We distinguish Bayesian Beta credible bounds from frequentist confidence intervals, characterize discounted/adaptively selected SPRT stopping honestly as an empirically calibrated sequential heuristic rather than claiming classical i.i.d. Wald bounds, and enforce a hard runtime capability boundary (`AgentEnvironmentInterface`) separating evaluator-only oracle state from all agent and policy modules.
- **Controlled Multi-Regime Evaluation & Execution-Channel Attribution:** Across six core experiments ($N = 20$ seeds), an extended $N = 30$ seed stability study, and a controlled 2×2 execution-channel fairness parity study, we separate statistical validation and precondition-gating gains from deterministic macro execution.
- **Transparent Reporting of Multiplicity, Trade-Offs, and Negative Boundaries:** Alongside large gains in unseen generalization (54.50%), continual learning (65.21%), post-shift recovery (+20.80 pp), and noisy-

feedback recovery (74.90%–84.10%), we export reproducible per-family (62/74 sig.) and simultaneous global-74 (60/74 sig.) Holm-Bonferroni ledgers and report all non-significant outcomes, high-noise false-rejection costs (65.44% FRR at $\eta = 0.35$), and short-horizon sample-complexity trade-offs without selective omission.

2 Motivation and Problem Statement

Consider an autonomous agent processing a sequential stream of structured multi-step tasks $\mathcal{S} = (q_1, q_2, \ldots, q_T)$ across domains $\mathcal{D}$ (e.g., API orchestration, system diagnosis, data pipeline repair, and resource planning). Each task $q_t = (x_t, d_t, \psi_t)$ presents a normalized surface query vector $x_t \in \mathbb{R}^m$, a domain identifier $d_t \in \mathcal{D}$, and an observable binary precondition vector $\psi_t \in \{0,1\}^K$. To solve q_t, the agent executes an H-step procedural workflow $\pi \in \mathcal{A}_0 \times \cdots \times \mathcal{A}_{H-1}$ within a per-episode interaction budget of $B \geq 1$ rollouts.

Let $c(q_t) \in \{0, \ldots, C-1\}$ denote a latent structural archetype determined by a subset of governing preconditions $G \subset \{0, \ldots, K-1\}$, while remaining bits $\{0, \ldots, K-1\} \setminus G$ represent nuisance context that varies across task instances without altering the optimal workflow $\pi^*_{d_t, c(q_t), r(t)}$ under active environment regime $r(t) \in \{0, 1, 2\}$. Crucially, neither $c(q_t)$, G, $r(t)$, nor π^* is ever observable to the agent; both `TaskInstance` and `RolloutResult` expose strictly observable inputs and noisy feedback. After each rollout π, the environment draws a true latent binary outcome $y_t \in \{0, 1\}$ governed by step alignment with $\pi^*_{d_t, c(q_t), r(t)}$, observed through a potentially corrupted feedback channel with symmetric flip probability $\eta_t \in [0, 0.5)$:

$$\tilde{y}_t = \begin{cases} y_t, & \text{with probability } 1 - \eta_t, \\ 1 - y_t, & \text{with probability } \eta_t. \end{cases} \tag{1}$$

Three coupled pathologies emerge in this regime when agents employ conventional episodic or workflow memory:

1. **Surface-Embedding Collision Pathology:** Tasks sharing identical surface topic tokens in x_t (yielding high cosine similarity $\cos(x_t, x_{t'}) > 0.85$) frequently differ on governing bits $\psi_{t,k} \neq \psi_{t',k}$ ($k \in G$). Embedding-only retrieval surfaces the previously successful plan $\pi_{t'}$, which mismatches required operators on q_t and suppresses exploratory discovery.
2. **Spurious & Corrupted Admission Pathology:** Even when π is suboptimal (e.g., matching $H-1$ of H steps with partial-alignment completion probability $p_{\text{sub3}} = 0.28$), or when $\eta_t > 0$ flips failures $y_t = 0$ into apparent successes $\tilde{y}_t = 1$, one-shot admission stores π permanently as a "verified" exemplar.
3. **Asymmetric Concept-Shift Pathology:** At unknown changepoint t_{shift}, a subset of domains $\mathcal{D}_{\text{shift}} \subset \mathcal{D}$ undergoes a concept shift $\pi^*_{d,c,0} \neq \pi^*_{d,c,1}$ while $\mathcal{D} \setminus \mathcal{D}_{\text{shift}}$ remains stationary. Agents without drift detection continue invoking obsolete plans on $\mathcal{D}_{\text{shift}}$, whereas naive global memory flushes destroy valid competence on $\mathcal{D} \setminus \mathcal{D}_{\text{shift}}$.

3 Related Work

Our work intersects four research threads: experiential memory for autonomous agents, memory reliability and poisoning defenses, sequential statistical inference and bandit confidence bounds, and online concept-drift adaptation. Table 1 summarizes how EVSA relates to representative frameworks without claiming unsupported "first" or "only" novelty on individual sub-mechanisms.

Experiential, Reflective, and Workflow Memory in Agents. ReAct [1] interleaves reasoning and action traces within a single episode but retains no cross-episode memory. Reflexion [2] and CLIN [17] persist verbal self-reflections or natural-language abstractions across trials, while Generative Agents [7] and MemGPT [18] manage long-horizon episodic stores via recency-relevance-importance scoring or OS-inspired paging. Moving from episodic traces to reusable skills, Voyager [3] stores executable code skills verified once in deterministic Minecraft tasks; ExpeL [4], Agent Workflow Memory (AWM) [5], Contextual Experience Replay (CER) [8], Dynamic Cheatsheet [19], and A-MEM [20] extract reusable workflows, evolving test-time cheatsheets, or linked Zettelkasten notes. Most recently, ReasoningBank [9] distills generalizable reasoning strategies from self-judged successful and failed trajectories paired with Memory-aware Test-Time Scaling (MaTTS), AdaMEM [21] combines offline long-term

Table 1: Architectural comparison of representative agentic memory and continual adaptation frameworks.

Framework / System	Cross-Task Memory	Multi-Trial Stat. Gate	Calibrated Reliability	Explicit Preconditions	Online Drift Detector	Selective Quarantine
ReAct [1]	No	No	No	No	No	No
Reflexion [2] / CLIN [17]	Verbal / Abstractions	No (1-episode)	Verbalized only	Implicit in text	No	Overwrite / Buffer cap
Voyager [3]	Code Skills	Single-trial env check	Binary pass/fail	Implicit in description	No	No
ExpeL [4] / AWM [5]	Insights / Workflows	Heuristic count/LLM	Empirical frequency	Prompt text similarity	No	Rule edit / Append
CER [8] / ReasoningBank [9]	Trajectories / Strategies	No (self-judged)	Self-judged tag	Dense embedding	No	Top-k dilution
Dynamic Cheatsheet [19] / A-MEM [20]	Curated Notes	LLM-judged edit	Uncalibrated LLM	Linked keywords	No	Prompt rewrite
AdaMEM [21]	Hybrid LT + ST Bank	LLM synthesis / MFT	Uncalibrated LLM	Query / state context	Short-term refresh	Summary overwrite
EVSA (This Study)	**3-Tier Strategies**	**Discounted SPRT Gate**	**Beta Credible Bound**	**Conjunctive ϕ_h + Safe LGG**	**Per-Domain CUSUM**	**Evidence Quarantine**

trajectory memory with dynamically synthesized short-term strategy memory and step-wise memory fine-tuning (STEP-MFT), and ExpRL [22] bounds exploration during reasoning-driven experience selection. Unlike ReasoningBank and AdaMEM—which distill or synthesize strategies over unquarantined trajectory stores without multi-trial sequential admission tests or online change-point invalidation—EVSA couples multi-trial sequential hypothesis gating at *admission*, conjunctive Safe-LGG precondition gating at *retrieval*, and domain-scoped residual CUSUM *invalidation* under concept shift.

Memory Reliability, Calibration, and Poisoning. Xiong et al. [10] demonstrate that verbalized LLM confidence is systematically overconfident, making self-scored memory filtering unreliable. In a 2026 ACL study, Xiong et al. [11] show that memory-augmented agents exhibit strong "experience-following" bias whereby misaligned stored records propagate errors over time and advocate selective addition coupled with deletion. From an adversarial perspective, AgentPoison [12] and PoisonedRAG [13] demonstrate that injecting a handful of crafted records into an unvalidated RAG/agent store hijacks downstream decisions at $> 90\%$ attack success. EVSA addresses stochastic and noisy-evaluator contamination structurally by requiring repeated empirical verification before promotion and monitoring post-promotion predictive residuals.

Sequential Hypothesis Testing, Bayesian Bounds, and Anytime-Valid Inference. Wald's Sequential Probability Ratio Test (SPRT) [23] minimizes expected sample size for testing simple binary hypotheses $H_0 : p = p_0$ against $H_1 : p = p_1$ under i.i.d. sampling. In adaptive multi-armed bandits, Kaufmann et al. [24] establish asymptotic optimality of Bayesian Upper Confidence Bounds (Bayes-UCB) using posterior Beta quantiles. Modern sequential statistics has further developed non-asymptotic time-uniform confidence sequences [25] and game-theoretic e-processes / test martingales [26] that preserve Type-I error under arbitrary adaptive stopping, alongside Holm's sequentially rejective multiple-testing control [27]. As detailed in Section 6, EVSA adopts discounted Beta credible bounds and a discounted SPRT-motivated log-odds gate for computational simplicity under non-stationarity, while explicitly auditing empirical false-promotion rates rather than claiming classical i.i.d. Wald bounds.

Concept Drift Detection, Continual Learning, and Inductive Generalization. Page's cumulative sum (CUSUM) scheme [16], adaptive windowing (ADWIN) [28], and drift-adaptation taxonomies surveyed by Gama et al. [14] detect distributional shifts in data streams via predictive error monitoring. In parametric continual learning and test-time adaptation, Elastic Weight Consolidation (EWC) [6], Prioritized Experience Replay (PER) [29], and Tent [30] mitigate forgetting or adapt normalization statistics online. For symbolic precondition induction, Plotkin's Least General Generalization (LGG) [15] computes maximal common conjunctions across positive instances. EVSA unifies Plotkin-style conservative precondition intersection with re-verification discounting and per-domain residual CUSUM quarantine inside a non-parametric agent strategy store.

4 Research Questions and Hypotheses

We structure our empirical investigation around five pre-specified Research Questions (**RQ1**–**RQ5**) and corresponding falsifiable null/alternative hypothesis pairs ($H_{0,k}$ vs. $H_{1,k}$) tested across paired random seeds:

- **RQ1 (Unseen Task Generalization & Execution-Channel Attribution):** Does coupling sequential statistical validation with precondition-gated retrieval improve held-out task success (executing strictly `VALIDATED` strategies) over memoryless and conventional memory baselines under identical rollout budgets, and does this gain persist when execution-channel adherence is equalized? *Hypotheses* $H_{0,1}/H_{1,1}$*:* Mean held-out success rate $\mu_{\text{EVSA}} = \mu_b$ vs. $\mu_{\text{EVSA}} \neq \mu_b$ across $b \in \{\text{Static, ConvMem, Reflection, Replay}\}$ and under matched `stochastic_prompt` vs. `direct_macro` execution.

- **RQ2 (Continual Learning & Memory Compactness):** Across multi-hundred-episode continual streams without manual resets, does EVSA exhibit monotonic competence accumulation without early negative transfer while maintaining a compact active strategy repository? *Hypotheses* $H_{0,2}/H_{1,2}$*:* Early Block-1 and terminal Block-8 probe success satisfy $\mu_{\text{EVSA}} = \mu_b$ vs. $\mu_{\text{EVSA}} \neq \mu_b$.
- **RQ3 (Component Necessity & Horizon Trade-Offs via Ablation):** What is the isolated contribution and sample-efficiency trade-off of each EVSA mechanism—sequential validation, Bayesian credible scoring, applicability preconditions, residual CUSUM drift detection, and selective memory invalidation—under noisy non-stationary streams? *Hypotheses* $H_{0,3}/H_{1,3}$*:* Removing module m yields zero paired change ($\mu_{\text{Full}} = \mu_{-m}$) vs. $\mu_{\text{Full}} \neq \mu_{-m}$.
- **RQ4 (Post-Shift Recovery & Selective Stationary Retention):** Following abrupt partial-domain concept shifts and compound shifts, does domain-scoped CUSUM invalidation accelerate post-shock adaptation without degrading unshifted stationary domains? *Hypotheses* $H_{0,4}/H_{1,4}$*:* Post-shift recovered SR, changepoint Adaptation Gain, proportional Recovery Rate, and unshifted-domain SR satisfy $\mu_{\text{EVSA}} = \mu_b$ vs. $\mu_{\text{EVSA}} \neq \mu_b$.
- **RQ5 (Feedback-Corruption Robustness & Overhead Trade-Offs):** Under symmetric evaluator noise $\eta \in \{0.10, 0.25, 0.35\}$, how effectively does sequential gating with Safe LGG suppress false acceptance and enable clean recovery, what False Rejection Rate (FRR) and False Discovery Proportion (FDP) trade-offs arise, and what computational overhead is incurred? *Hypotheses* $H_{0,5}/H_{1,5}$*:* Active memory contamination and post-corruption clean recovery satisfy $\mu_{\text{EVSA}} = \mu_b$ vs. $\mu_{\text{EVSA}} \neq \mu_b$.

5 Proposed EVSA Architecture

EVSA organizes experiential adaptation as a ten-module closed-loop architecture (Figure 1). Rather than storing unstructured transcripts, the **Tiered Strategy Repository** (`src/memory/repository.py`) maintains structured hypotheses $h = (\text{id}_h, d_h, c_h, \phi_h, \pi_h, \tau_h, \alpha_h, \beta_h, \Lambda_h, n_h)$, where $d_h \in \mathcal{D}$ is the task domain, $c_h \in \mathbb{R}^m$ is the unit-normalized query centroid, $\phi_h \subseteq \{0, \dots, K-1\} \times \{0, 1\}$ is a conjunctive precondition set, $\pi_h \in \mathcal{A}_0 \times \cdots \times \mathcal{A}_{H-1}$ is an H-step procedural workflow, and τ_h denotes the epistemic tier in $\{\texttt{CANDIDATE}, \texttt{VALIDATED}, \texttt{QUARANTINED}\}$ (Figure 2).

Operational Episode Loop. Algorithm 1 details EVSA's per-episode decision and update cycle (`src/agents/evsa_agent.py`):

1. **Exploitation via Validated Retrieval (Modules 3 & 7):** Given task $q_t = (x_t, d_t, \psi_t)$, the **Applicability Retriever** filters `VALIDATED` strategies in domain d_t whose precondition conjunction $\phi_h(\psi_t) = 1$ holds and whose Bayesian lower credible bound satisfies $\underline{p}_h^{\text{cred}}(\delta) \geq \tau_{\text{rel}} = 0.45$ (Figure 3a). If a match h^* exists, EVSA records the out-of-sample predictive probability $\hat{p}_{\text{pre}} = \alpha_{h^*}/(\alpha_{h^*} + \beta_{h^*})$, executes π_{h^*}, updates $(\alpha_{h^*}, \beta_{h^*}, \Lambda_{h^*})$ first with observed feedback $\tilde{y}_t$, and then feeds residual $e_t = \hat{p}_{\text{pre}} - \tilde{y}_t$ to the domain-d_t **Residual CUSUM Detector** (Module 9).
2. **Active Candidate Verification During Adaptation Only (Modules 4–6):** During online learning (`learn` = **true**), if no `VALIDATED` strategy applies (or if $\tilde{y}_t = 0$ leaves rollout budget remaining), EVSA selects among precondition-compatible `CANDIDATE` strategies maximizing a Bayes-UCB utility ($0.85\, I^{-1}_{0.90}(\alpha_h, \beta_h) + 0.15 \max(0, \cos(x_t, c_h)) + 0.015|\phi_h|$) [24]. During frozen evaluation (`learn` = **false**), unvalidated `CANDIDATE` strategies are strictly excluded from execution.
3. **Salience-Guided Exploration & Safe LGG Admission (Modules 2 & 8):** When budget remains and no strategy is selected, the **Hypothesis Generator** synthesizes a workflow π_{new} (mutating away from tombstoned plans in $\mathcal{M}_{d_t}$ matching ψ_t) using marginal step-predicate outcome contrasts accumulated strictly from observed rollouts. Only if $\tilde{y}_t = 1$ is π_{new} admitted to `CANDIDATE`—or merged into an existing strategy matching (d_t, π_{new}) via Safe LGG precondition intersection.
4. **Selective Drift Quarantine & Capacity Pruning (Modules 8 & 9):** If domain-d_t's CUSUM statistic crosses h_{drift} (subject to post-reset warmup and cooldown), EVSA immediately quarantines active strategies in $\mathcal{M}_{d_t}$ exhibiting empirical failure ($f_h^{\text{consec}} \geq 1$) or weakened reliability ($\underline{p}_h^{\text{cred}} < 0.58$), while discounting unfalsified strategies in $\mathcal{M}_{d_t}$ back to `CANDIDATE` for rapid re-verification and leaving all other domains untouched.

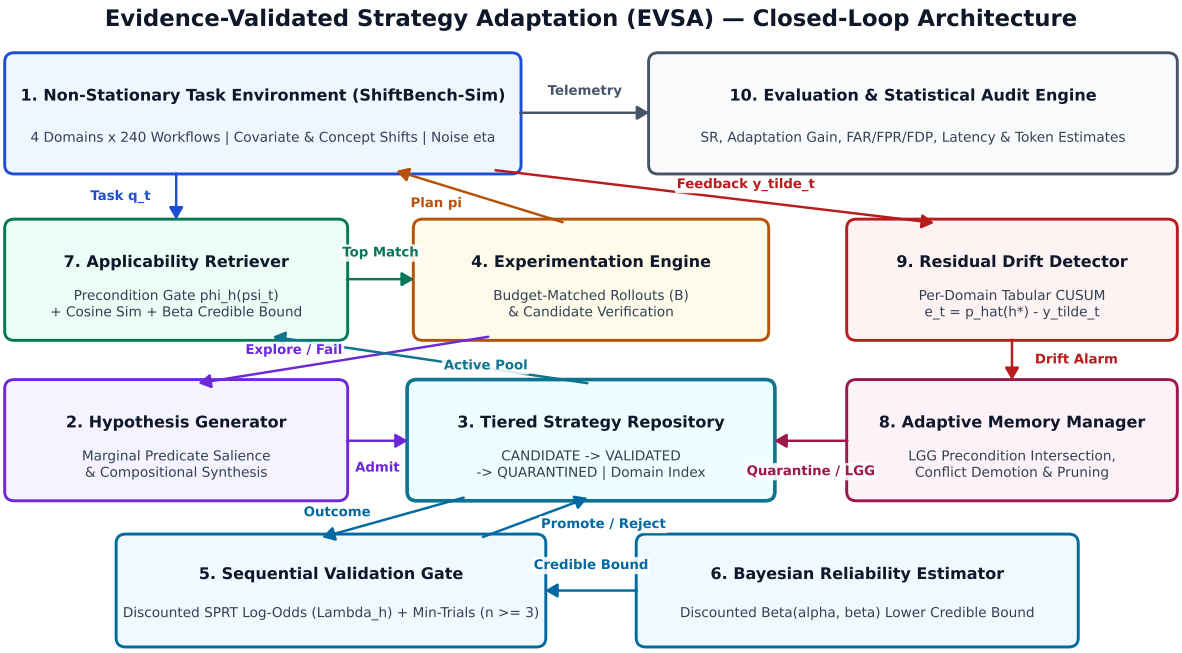


Figure 1: EVSA Ten-Module Closed-Loop System Architecture. Tasks q_t are matched against precondition- and reliability-gated strategies (Module 7), executed or verified within rollout budget B (Module 4), audited sequentially via discounted SPRT log-odds and Beta lower credible bounds (Modules 5–6), generalized conservatively via Safe LGG (Module 8), and monitored online per domain by a residual tabular CUSUM drift detector (Module 9).

6 Mathematical Formulation

6.1 Discounted Bayesian Beta Posterior and Lower Credible Bounds

Let $\theta_h \in (0,1)$ denote the unknown conditional success rate of strategy h on tasks satisfying $\phi_h(\psi_t) = 1$ under the active feedback channel. Placing a uniform prior $\theta_h \sim \text{Beta}(\alpha_0, \beta_0)$ ($\alpha_0 = \beta_0 = 1$), we incorporate exponential forgetting factor $\gamma = 0.985$ upon observing outcome $\tilde{y}_n \in \{0,1\}$ (`src/validation/confidence.py`):

$$\alpha_h^{(n)} = \alpha_0 + \gamma\big(\alpha_h^{(n-1)} - \alpha_0\big) + \tilde{y}_n, \qquad \beta_h^{(n)} = \beta_0 + \gamma\big(\beta_h^{(n-1)} - \beta_0\big) + (1 - \tilde{y}_n). \tag{2}$$

The posterior predictive mean is $\hat{p}_h = \alpha_h^{(n)}/(\alpha_h^{(n)} + \beta_h^{(n)})$, and the conservative reliability score is the exact δ-quantile ($\delta = 0.10$) of $\text{Beta}(\alpha_h^{(n)}, \beta_h^{(n)})$ via the inverse regularized incomplete beta function I_δ^{-1} (with symmetric upper quantile $\bar{p}_h^{\text{cred}} = I_{1-\delta}^{-1}$ for Bayes-UCB candidate exploration):

$$\underline{p}_h^{\text{cred}}(\delta) = I_\delta^{-1}\big(\alpha_h^{(n)}, \beta_h^{(n)}\big), \qquad \text{satisfying} \int_0^{\underline{p}_h^{\text{cred}}(\delta)} \frac{u^{\alpha_h^{(n)}-1}(1-u)^{\beta_h^{(n)}-1}}{\text{B}(\alpha_h^{(n)}, \beta_h^{(n)})}\, du = \delta. \tag{3}$$

Credible Bounds vs. Unregularized Empirical Ablation: Crucially, $\underline{p}_h^{\text{cred}}(\delta)$ is a *discounted Bayesian* $(1-\delta)$ *lower credible bound*, distinct from a frequentist Clopper-Pearson confidence bound or time-uniform confidence sequence [25]. Under small sample sizes, $\underline{p}_h^{\text{cred}}(0.10)$ regularizes 1-for-1 flukes ($\alpha = 2, \beta = 1 \Rightarrow 0.316 < 0.48$) while $\bar{p}_h^{\text{cred}}(0.90)$ protects promising candidates against single stochastic misses. In our `w/o Confidence Estimation` ablation, Bayesian credible bounds are replaced by the unregularized discounted empirical frequency $\hat{p}_h^{\text{emp}} = (\alpha_h - \alpha_0)/(\alpha_h - \alpha_0 + \beta_h - \beta_0)$ (0.50 at $n_h = 0$).

6.2 Discounted Log-Odds Sequential Heuristic (Motivated by Wald's SPRT)

To separate aligned strategies ($\theta_h \geq p_1$) from suboptimal partial plans ($\theta_h \leq p_0$), we specify indifference boundaries $p_0 = 0.35$ and $p_1 = 0.80$ with nominal calibration parameters $(\alpha_{\text{FA}}, \beta_{\text{FR}}) = (0.11, 0.15)$

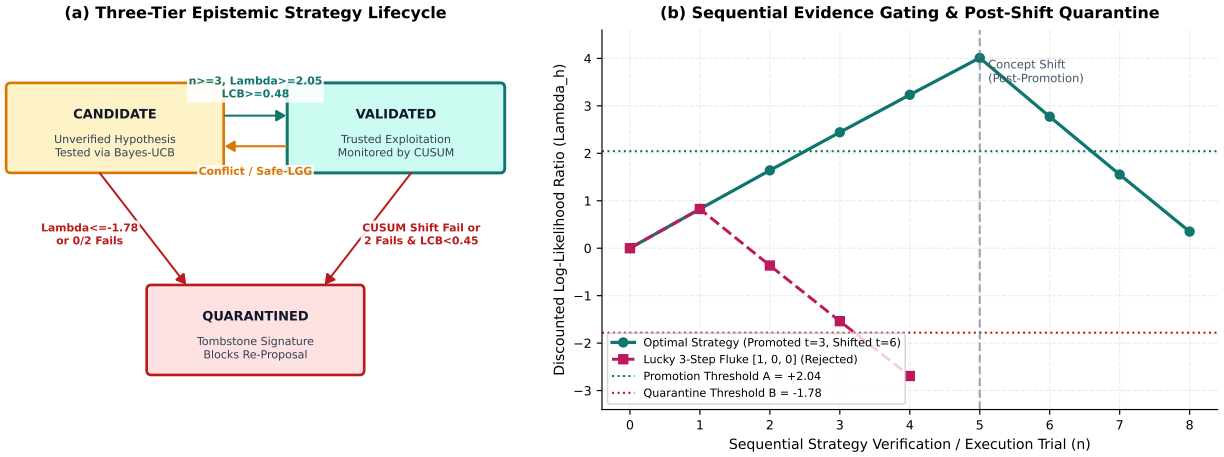


Figure 2: Strategy Validation Lifecycle and Sequential Evidence Trajectories. (a) State transitions across `CANDIDATE`, `VALIDATED`, and `QUARANTINED` tiers, including conflict/Safe-LGG demotion and tombstone re-admission blocking. (b) Exact deterministic trajectories of the discounted log-likelihood ratio Λ_h ($\gamma = 0.985$) showing promotion of a true optimal workflow at $n = 3$, rejection of a 1-success lucky fluke $[1, 0, 0]$, and rapid post-shift decay.

(`src/validation/sprt_validator.py`), yielding log-odds thresholds:

$$A = \ln\left(\frac{1-\beta_{\text{FR}}}{\alpha_{\text{FA}}}\right) \approx +2.045, \qquad B_{\text{low}} = \ln\left(\frac{\beta_{\text{FR}}}{1-\alpha_{\text{FA}}}\right) \approx -1.780. \tag{4}$$

Upon each trial $\tilde{y}_n \in \{0, 1\}$, EVSA updates a discounted, clipped log-likelihood ratio $\Lambda_h^{(n)} \in [-\Lambda_{\max}, +\Lambda_{\max}]$ ($\Lambda_{\max} = 6.0$):

$$\Lambda_h^{(n)} = \text{clip}\left(\gamma\Lambda_h^{(n-1)} + \tilde{y}_n \ln\left(\frac{p_1}{p_0}\right) + (1-\tilde{y}_n)\ln\left(\frac{1-p_1}{1-p_0}\right),\ -\Lambda_{\max},\ +\Lambda_{\max}\right), \tag{5}$$

where $\ln(p_1/p_0) \approx +0.827$ and $\ln((1-p_1)/(1-p_0)) \approx -1.179$. Tier transitions follow the conjunctive decision rule:

$$\tau_h^{(n)} = \begin{cases} \texttt{VALIDATED}, & \text{if } \tau_h^{(n-1)} = \texttt{CANDIDATE} \wedge n_h \geq n_{\min} \wedge \Lambda_h^{(n)} \geq A \wedge \underline{p}_h^{\text{cred}}(0.10) \geq 0.48, \\ \texttt{QUARANTINED}, & \text{if } \Lambda_h^{(n)} \leq B_{\text{low}} \ \vee \ (\tau_h^{(n-1)} = \texttt{CANDIDATE} \wedge n_h \geq 2 \wedge s_h = 0) \\ & \quad \vee \ (\tau_h^{(n-1)} = \texttt{VALIDATED} \wedge f_h^{\text{consec}} \geq 2 \wedge \underline{p}_h^{\text{cred}}(0.10) < 0.45), \\ \tau_h^{(n-1)}, & \text{otherwise,} \end{cases} \tag{6}$$

with $n_{\min} = 3$, cumulative successes s_h, and consecutive failures f_h^{consec}. Three consecutive successes produce $\Lambda_h^{(3)} \approx 2.443 \geq 2.045$, promoting h cleanly.

Theoretical Audit of Classical SPRT Assumptions: Classical Wald (1945) guarantees ($\Pr(\text{Accept } H_1 \mid H_0) \leq \alpha_{\text{FA}}$) require undiscounted ($\gamma = 1$), unclipped i.i.d. sampling independent of arm selection [23]. Because Eq. (5) introduces exponential discounting ($\gamma < 1$), clipping ($\pm\Lambda_{\max}$), a hard sample-size floor ($n_{\min} = 3$), a joint Beta credible gate, and adaptive Bayes-UCB candidate selection, *classical Wald Type-I/II error bounds do not hold analytically.* We therefore characterize Eqs. (4)–(6) strictly as an empirically calibrated sequential heuristic and measure its realized candidate-level false-promotion rate directly in Experiment 4 (8.08%–11.32% across $\eta \in [0.10, 0.35]$). Achieving exact finite-sample error guarantees under adaptive bandit stopping requires non-negative test supermartingales / e-processes [26].

6.3 Applicability-Gated Retrieval and Re-Verified Safe LGG Abstraction

Each strategy h stores a conjunctive precondition $\phi_h \subseteq \{0, \ldots, K-1\} \times \{0, 1\}$ evaluated via $\phi_h(\psi_t) = \prod_{(k,v)\in\phi_h} \mathbb{I}[\psi_{t,k} = v] \in \{0, 1\}$. When an exploratory workflow π succeeds ($\tilde{y}_t = 1$) with induced preconditions ϕ_{new} and matches an active peer $h \in \mathcal{M}_{d_t}$ ($\pi_h = \pi$) differing on $1 \leq \Delta \leq 2$ shared coordinates, EVSA applies Plotkin's Least General Generalization (LGG) [15] (`src/memory/memory_manager.py`):

$$\phi_h^{\text{LGG}} = \left\{(k, v) \in \phi_h \cap \phi_{\text{new}}\right\}, \qquad \text{subject to } |\phi_h^{\text{LGG}}| \geq K_{\min} = 3. \tag{7}$$

Algorithm 1 EVSA Closed-Loop Decision, Safe-LGG Validation, and Selective Drift Quarantine

Require: Task $q_t = (x_t, d_t, \psi_t)$, Strategy Repository $\mathcal{M}$, Budget B, Mode $\texttt{learn} \in \{\textbf{true}, \textbf{false}\}$
$b \leftarrow 0$, tried $\leftarrow \emptyset$
while $b < B$ **do**
  $\mathcal{V}_t \leftarrow \{h \in \mathcal{M}_{d_t} \setminus \text{tried} : \tau_h = \texttt{VALIDATED} \wedge \phi_h(\psi_t) = 1 \wedge \underline{p}_h^{\text{cred}}(\delta) \geq \tau_{\text{rel}}\}$
  $\mathcal{C}_t \leftarrow \{h \in \mathcal{M}_{d_t} \setminus \text{tried} : \tau_h = \texttt{CANDIDATE} \wedge \phi_h(\psi_t) = 1\}$
  if $\mathcal{V}_t \neq \emptyset$ **or** ($\texttt{learn}$ **and** $\mathcal{C}_t \neq \emptyset$) **then**
    Select h^* maximizing $U(h, q_t)$ on $\mathcal{V}_t$ if $\mathcal{V}_t \neq \emptyset$, else maximizing $U_{\text{cand}}(h, q_t)$ on $\mathcal{C}_t$
    used_val $\leftarrow (\tau_{h^*} = \texttt{VALIDATED})$; $\hat{p}_{\text{pre}} \leftarrow \text{PosteriorMean}(h^*)$; tried $\leftarrow$ tried $\cup \{h^*\}$
    Execute π_{h^*}, observe $\tilde{y}_t \in \{0, 1\}$; $b \leftarrow b + 1$
    if $\texttt{learn}$ **then**
      Update $(n_{h^*}, \alpha_{h^*}, \beta_{h^*}, f_{h^*}^{\text{consec}}, \Lambda_{h^*}, \tau_{h^*})$ via Eqs. (2)–(6)
      if used_val **and** UpdateDomainCUSUM$(d_t, \hat{p}_{\text{pre}} - \tilde{y}_t, t) = \texttt{ALARM}$ **then**
        Quarantine $h \in \mathcal{M}_{d_t}$ with $f_h^{\text{consec}} \geq 1 \vee \underline{p}_h^{\text{cred}} < 0.58$; discount unfalsified $h \in \mathcal{M}_{d_t}$ to $\texttt{CANDIDATE}$
      end if
    end if
  else
    Synthesize non-quarantined plan π_{exp} via observed step-predicate salience; execute π_{exp}, observe $\tilde{y}_t$; $b \leftarrow b + 1$
    if $\texttt{learn}$ **and** $\tilde{y}_t = 1$ **then**
      Merge via Safe LGG (Eq. (7)) if compatible peer $h \in \mathcal{M}_{d_t}$ shares π_{exp}; else admit new $\texttt{CANDIDATE}$
    end if
  end if
  if $\tilde{y}_t = 1$ **then break**
  end if
end while

Safe LGG Re-Verification: Whenever Eq. (7) strictly broadens an already-$\texttt{VALIDATED}$ strategy ($|\phi_h^{\text{LGG}}| < |\phi_h|$), preserving $\texttt{VALIDATED}$ status blindly would allow a single noisy false positive ($\tilde{y}_t = 1$) on a neighboring archetype to drop a governing literal and poison future exploitations. EVSA therefore discounts excess pseudo-counts by $\gamma_{\text{LGG}} = 0.80$, clamps $\Lambda_h \leftarrow \min(\gamma_{\text{LGG}}\Lambda_h, A - 0.15)$, and steps τ_h back to $\texttt{CANDIDATE}$ so that at least one independent confirming trial on the generalized subspace is required before re-promotion.

Retrieval utility enforces both symbolic applicability $\phi_h(\psi_t) = 1$ and minimum lower credible reliability $\tau_{\text{rel}} = 0.45$ for $\texttt{VALIDATED}$ strategies ($w_{\text{sem}} = 0.28, w_{\text{conf}} = 0.52, w_{\text{tier}} = 0.20, b_{\text{spec}} = 0.015$; `src/retrieval/retriever.py`):

$$U(h, q_t) = \mathbb{I}[d_h = d_t] \cdot \phi_h(\psi_t) \cdot \mathbb{I}\big[\underline{p}_h^{\text{cred}}(0.10) \geq \tau_{\text{rel}}\big] \cdot \Big(w_{\text{sem}} \max(0, \cos(x_t, c_h)) + w_{\text{conf}} \underline{p}_h^{\text{cred}}(0.10) + w_{\text{tier}} + b_{\text{spec}} |\phi_h|\Big). \tag{8}$$

6.4 Per-Domain Residual Tabular CUSUM Drift Detection

Let $e_t = \hat{p}_{h^*}^{(t-1)} - \tilde{y}_t \in [-1, 1]$ denote the one-step out-of-sample prediction residual computed *prior* to updating h^* whenever a $\texttt{VALIDATED}$ strategy h^* is executed in domain $d = d_t$ (`src/drift_detection/cusum_detector.py`). EVSA maintains an independent one-sided tabular CUSUM statistic [16] per domain $d \in \mathcal{D}$:

$$S_d^{(t)} = \max\big(0,\ S_d^{(t-1)} + e_t - \nu_{\text{drift}}\big), \qquad S_d^{(0)} = 0, \tag{9}$$

with slack $\nu_{\text{drift}} = 0.12$, alarm threshold $h_{\text{drift}} = 1.45$, post-reset observation warmup $n_{\text{min,obs}} = 3$, and per-domain cooldown of 12 episodes. Firing $S_d^{(t)} \geq h_{\text{drift}}$ immediately quarantines strategies in $\mathcal{M}_d$ exhibiting $f_h^{\text{consec}} \geq 1$ or $\underline{p}_h^{\text{cred}}(0.10) < 0.58$, discounts unfalsified strategies in $\mathcal{M}_d$ by $\gamma_{\text{shift}} = 0.35$ to $\texttt{CANDIDATE}$, and resets $S_d^{(t)} \leftarrow 0$ (Figure 4).

7 Benchmark Design: ShiftBench-Sim v1.0

Evaluating memory validation, drift invalidation, and noisy-feedback calibration across hundreds of thousands of rollouts requires exact control over ground-truth structural invariants without hidden pretraining leakage. We

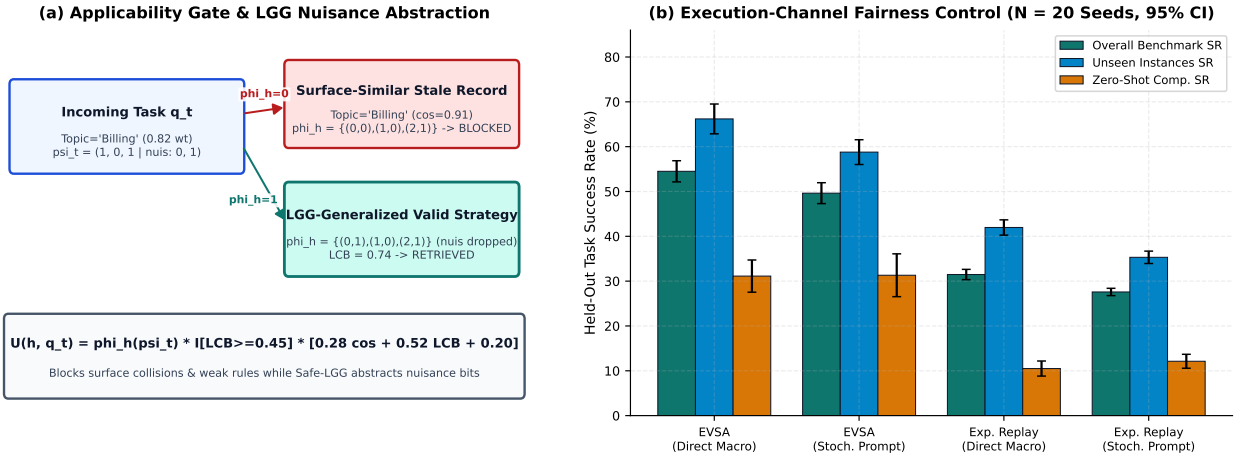


Figure 3: Applicability-Gated Retrieval and Execution-Channel Fairness Parity. (a) Multiplicative gating $\phi_h(\psi_t) \cdot \mathbb{I}[\underline{p}_h^{\text{cred}} \geq 0.45]$ blocks surface-similar stale records ($\cos = 0.91$) while retrieving Safe-LGG generalized valid strategies. (b) Controlled $N = 20$ seed fairness experiment decoupling memory-selection quality from execution-channel determinism (`Direct Macro` vs. `Stochastic Prompt`).

construct **ShiftBench-Sim v1.0** (`src/benchmark/environment.py`), a controlled, low-dimensional synthetic symbolic-procedural multi-domain simulation (not a live neural LLM or natural-language benchmark) parameterized as follows:

- **Domains and Heterogeneous Action Space:** $|\mathcal{D}| = 4$ operational domains (`api_orchestration`, `system_diagnosis`, `data_pipeline`, `resource_planning`). Each task requires planning an $H = 4$-step discrete operator workflow $\pi \in \mathcal{A}_0 \times \mathcal{A}_1 \times \mathcal{A}_2 \times \mathcal{A}_3$ with action cardinalities $(4, 5, 4, 3)$ (240 distinct workflows per domain).
- **Governing vs. Nuisance Preconditions:** Each task exposes $K = 5$ observable binary predicates $\psi_t \in \{0,1\}^5$. Bits (ψ_0, ψ_1, ψ_2) define $C = 2^3 = 8$ latent workflow archetypes $c = 4\psi_0 + 2\psi_1 + \psi_2 \in \{0, \ldots, 7\}$. Bits (ψ_3, ψ_4) are pure nuisance attributes that alter surface context without changing π^*. Archetypes $c \in \{0, \ldots, 6\}$ appear during stationary training; **archetype $c = 7$ ($\psi_0 = \psi_1 = \psi_2 = 1$) is held out strictly for zero-shot compositional testing**, with Steps 0–2 governed additively by individual bits and Step 3 governed by a non-linear XOR parity interaction $(\psi_0 \oplus \psi_1)$.
- **Controlled Surface-Embedding Collisions:** Normalized synthetic task embeddings $x_t \in \mathbb{R}^{20}$ ($\|x_t\|_2 = 1$) concatenate domain features (weight 0.50), noisy structural features (weight 0.36), and 8 shared surface-topic vectors (weight 0.82). Tasks sharing surface topics across conflicting archetypes therefore exhibit high cosine similarity, penalizing precondition-blind retrieval.
- **Regime Shifts and Stochastic Outcome Probabilities:** Regime $r = 0$ defines baseline optimal workflows $\pi^*_{d,c,0}$. Regime $r = 1$ (abrupt partial-domain concept shift) shifts Steps 1 and 2 by $+2$ (mod $|\mathcal{A}_s|$) across three domains while keeping `resource_planning` invariant ($r = 0$) as an unshifted control. Regime $r = 2$ (compound shift) shifts Step 0 and Step 2 across all four domains alongside covariate concentration onto $c \in \{4, 5, 6, 7\}$. Executing 4, 3, 2, or ≤ 1 matching steps succeeds with probabilities $(p_{\text{opt}}, p_{\text{sub3}}, p_{\text{sub2}}, p_{\text{sub1}}) = (0.92, 0.28, 0.09, 0.02)$.
- **Hard Capability-Isolated Oracle Boundary:** Beyond omitting oracle fields from `TaskInstance`, `RolloutResult`, and `AgentEpisodeOutcome`, all agent rollouts are routed through `AgentEnvironmentInterface` (`src/benchmark/agent_interface.py`), a slot-restricted capability wrapper exposing solely `evaluate_plan(task, executed_plan)` while evaluator-side telemetry (`EpisodeTelemetry`) is finalized exclusively inside `run_audited_episode` (`src/evaluation/audit.py`). Passing a raw `NonStationaryTaskEnvironment` to any agent immediately raises `TypeError`, and `CalibratedSymbolicLLMBackbone` (`src/benchmark/llm_backbone.py`) computes zero-shot step priors (0.67 marginal step competence) strictly from observable (d_t, ψ_t) using a linear inclusive-OR heuristic $(\psi_0 \vee \psi_1)$ on Step 3—verified programmatically by `test_hard_interface_boundary_separates_agent_from_oracle` and `test_zero_oracle_leakage_across_all_agent_and_backbone_modules`.

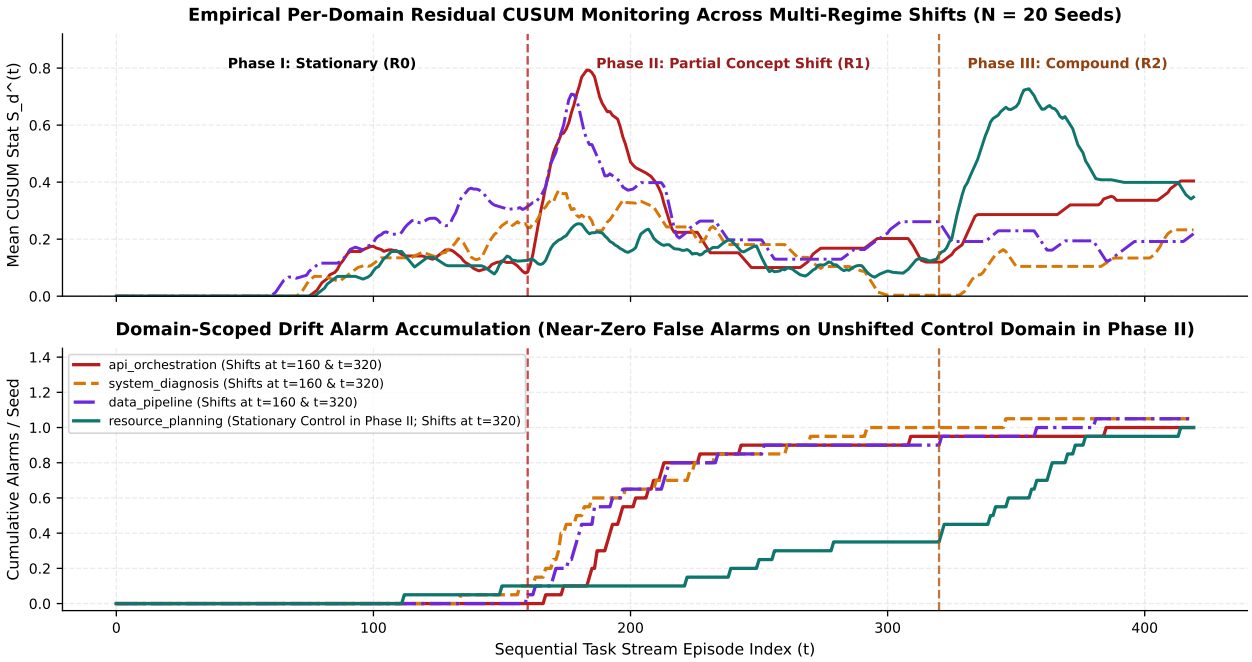


Figure 4: Empirical Per-Domain Residual CUSUM Monitoring Across Multi-Regime Shifts ($N = 20$ Seeds). Top: 5-episode smoothed mean CUSUM statistic $S_d^{(t)}$ across 420 episodes in Experiment 3. Bottom: Cumulative drift alarms per seed. At $t = 160$, the three shifted domains spike and trigger quarantine within a mean delay of 38.40 global episodes (≈ 9.6 domain episodes), whereas the stationary control domain (`resource_planning`) averages only 0.25 false alarms across Phase II before genuinely shifting at $t = 320$.

8 Experimental Methodology

All agents share the identical `CalibratedSymbolicLLMBackbone`, identical task streams per seed, and identical per-episode rollout budget B ($B = 2$ during adaptation, $B = 1$ during frozen evaluation). Six core experiments plus two sensitivity audits are executed across $N = 20$ primary seeds ($\{1001, \ldots, 1020\}$; extended to $N = 30$ seeds $\{1001, \ldots, 1030\}$ for stability verification):

1. **Experiment 1 (Unseen Task Generalization):** Online adaptation on 240 stationary tasks ($r = 0, \eta = 0, B = 2$, archetypes $0 \ldots 6$), followed by frozen evaluation (`learn` = **false**, $B = 1$) on 240 unseen tasks (160 unseen surface instances of $c \in \{0 \ldots 6\}$ and 80 zero-shot compositional tasks from held-out $c = 7$).
2. **Experiment 2 (Continual Learning Without Resets):** 320 sequential training episodes divided into 8 blocks of 40 tasks ($r = 0, \eta = 0, B = 2$), probed ($B = 1$) after every block (plus $t = 0$) on an independent 96-task held-out suite.
3. **Experiment 3 (Multi-Regime Distribution Shift):** 420-episode continuous stream ($B = 2, \eta = 0$): Phase I ($t \in [0, 160)$, stationary $r = 0$), Phase II ($t \in [160, 320)$, abrupt concept shift $r = 1$ in 3 domains while `resource_planning` remains stationary), and Phase III ($t \in [320, 420)$, compound shift $r = 2$ across all 4 domains over $c \in \{4, 5, 6, 7\}$).
4. **Experiment 4 (Misleading & Noisy Feedback Robustness):** Across symmetric corruption rates $\eta \in \{0.10, 0.25, 0.35\}$, agents train for 200 noisy episodes followed by 100 clean episodes ($\eta = 0$) to measure Rollout FAR, Candidate FPR, FRR, Active Validated Contamination (FDP), and Clean Recovery SR.
5. **Experiment 5 (Leave-One-Out Component Ablation):** Evaluates Full EVSA against five single-module ablations on a 360-episode stream combining pre-shift noisy feedback ($\eta = 0.15, t \in [0, 180)$) with an abrupt 3-domain concept shift ($r = 1, \eta = 0, t \in [180, 360)$) plus a 96-task unseen Regime-1 evaluation suite.
6. **Experiment 6 (Computational Efficiency & Projected Token Cost):** Records local wall-clock CPU latency (`time.perf_counter`), memory footprint (`KB`), environment rollouts, and analytically projected token/API costs (modeled at \$2.50/\$10.00 per 10^6 input/output tokens; \$0.00 actual commercial API spend).

Table 2: Experiment 1: Held-out unseen benchmark generalization ($N = 20$ seeds, 240 tasks/seed; Mean ± SD and [95% CI]).

Agent Architecture	Overall SR (%)	Exact Plan Acc. (%)	Unseen Instances SR (%)	Zero-Shot Comp. $c = 7$ (%)	Avg. Reward
Baseline A: Static	26.48 ± 3.04 [25.06, 27.90]	12.38 ± 2.46	30.75 ± 3.78	17.94 ± 4.61	0.386 ± 0.023
Baseline B: Conventional Memory	29.62 ± 2.77 [28.33, 30.92]	19.60 ± 2.51	37.81 ± 3.39	13.25 ± 2.82	0.410 ± 0.021
Baseline C: Reflection	25.65 ± 3.61 [23.96, 27.33]	13.56 ± 3.23	29.81 ± 4.08	17.31 ± 5.88	0.377 ± 0.030
Baseline D: Experience Replay	27.58 ± 1.74 [26.77, 28.40]	15.69 ± 1.82	35.31 ± 2.96	12.12 ± 3.32	0.389 ± 0.016
Proposed Agent E: EVSA	$\mathbf{54.50 \pm 5.04}$ [**52.14, 56.86**]	$\mathbf{45.40 \pm 7.01}$	$\mathbf{66.19 \pm 7.11}$	$\mathbf{31.13 \pm 7.67}$	$\mathbf{0.646 \pm 0.041}$

9 Baseline Implementations and Fairness Audit

To ensure budget-matched algorithmic comparisons on ShiftBench-Sim v1.0, we implement four symbolic-procedural baseline adaptations in `src/agents/baselines.py`—explicitly distinguishing these controlled benchmark adaptations from exact natural-language reproductions of live LLM systems:

- **Baseline A: Static Agent** (*ReAct-style memoryless adaptation* [1]): Zero cross-episode memory; samples workflows strictly from `CalibratedSymbolicLLMBackbone` within rollout budget B.
- **Baseline B: Conventional Memory Agent** (*Generative-Agents-style episodic buffer* [7]): Stores up to 400 episodic records scored by $0.55 \cos(x_t, x_i) + 0.20 \cdot 0.993^{\Delta t} + 0.25\, \tilde{r}_i$, conditioning generation on top-3 positive matches (weighted adherence $\in [0.55, 0.96]$).
- **Baseline C: Reflection Agent** (*Reflexion-style sliding critique adaptation* [2]): Maintains per-domain sliding buffers (10/domain) of step-failure avoidance constraints and successful workflows without multi-trial statistical gating or drift quarantine.
- **Baseline D: Experience Replay Agent** (*CER/AWM-style distilled workflow buffer* [5, 8]): Retains successful ($\tilde{y}_t = 1$) workflows (200 max, centroid-deduplicated at $\cos > 0.80$), retrieving top-3 domain matches above $\cos > 0.25$.

Audit of Execution-Channel Asymmetry and Controlled Parity Design. In default mode, once a strategy reaches `VALIDATED`, EVSA invokes π_h directly as a compiled deterministic macro, whereas Baselines B–D pass retrieved workflows through `backbone.execute_guided_plan` (weighted adherence ≤ 0.96). To prevent misattributing macro determinism to memory quality, we evaluate a full 2×2 execution-channel parity control (`Direct Macro` vs. `Stochastic Prompt` in both EVSA and Experience Replay) across $N = 20$ seeds (Figure 3b and Table 3).

10 Experimental Results

10.1 Experiment 1: Unseen Task Generalization and Execution Parity

Table 2 and Figure 5 present held-out generalization across $N = 20$ seeds (240 unseen tasks/seed, executing strictly `VALIDATED` strategies during frozen evaluation). EVSA achieves **54.50% ± 5.04%** overall task success (95% CI [52.14%, 56.86%], median 53.75%, exact plan accuracy 45.40%), outperforming *Static* (26.48%, +105.82% relative, $d_z = 4.51$), *Conventional Memory* (29.62%, +83.97%, $d_z = 4.47$), *Reflection* (25.65%, +112.51%, $d_z = 4.31$), and *Experience Replay* (27.58%, +97.58%, $d_z = 6.35$), with all four Holm-Bonferroni adjusted paired t-test p-values $\leq 8.15 \times 10^{-14}$. Extending Experiment 1 to $N = 30$ **independent seeds** confirms tight distributional stability: EVSA achieves **55.17% ± 5.62%** (95% CI [53.07%, 57.27%]) versus 25.18%–29.12% across baselines.

Sub-Split Decomposition & Compositional Parity Ceiling: On unseen surface instances of seen archetypes ($c \in \{0, \ldots, 6\}$), Conventional Memory (37.81%) and Experience Replay (35.31%) modestly improve over Static (30.75%), whereas EVSA reaches **66.19% ± 7.11%** by blocking surface-topic collisions and abstracting nuisance bits (ψ_3, ψ_4). On **zero-shot compositional archetype** $c = 7$ **($\psi_0 = \psi_1 = \psi_2 = 1$)**—where the non-oracle backbone's linear inclusive-OR heuristic predicts the wrong Step-3 operator—Conventional Memory (13.25%) and Experience Replay (12.12%) drop below Static (17.94%) due to surface collisions with seen archetypes. Because EVSA's precondition gate rejects colliding whole plans while its Hypothesis Generator composes validated step-level marginal operators across Steps 0, 1, and 2 (achieving 3-of-4 step alignment with $p_{\text{sub3}} = 0.28$), EVSA reaches **31.13% ± 7.67%** (+13.19 pp over Static, $d_z = 1.57, p_{\text{Holm}} = 1.07 \times 10^{-6}$; +19.00 pp over Replay,

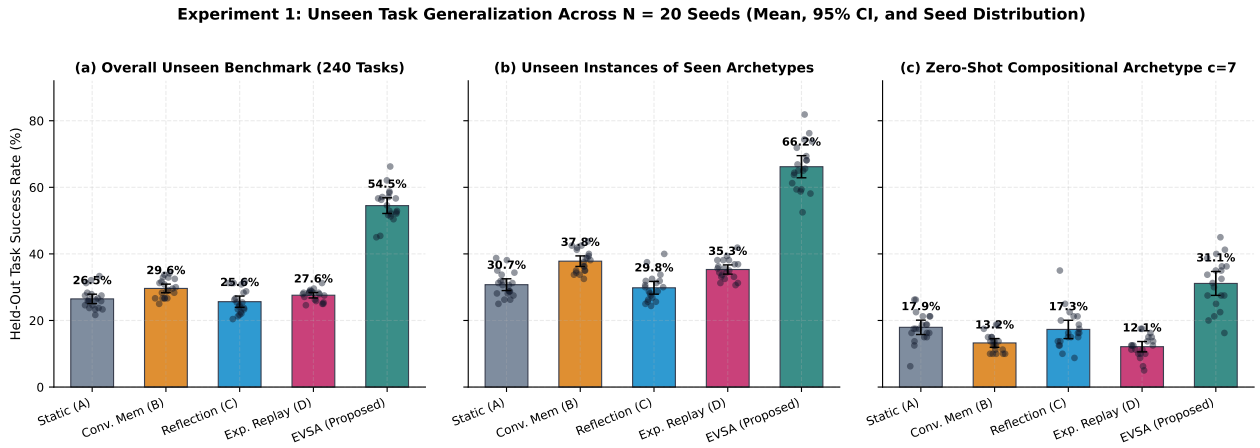


Figure 5: Experiment 1 Unseen Generalization Across Splits ($N = 20$ Seeds). Bars depict group means with 95% CIs; overlaid points show individual seed realizations. Memory baselines suffer negative transfer on zero-shot archetype $c = 7$ (12.1%–17.3% vs. Static 17.9%), whereas EVSA composes validated marginal operators on Steps 0–2 to reach 31.1% on $c = 7$ and 66.2% on unseen instances of seen archetypes.

Table 3: Execution-Channel Fairness Parity Control on Experiment 1 ($N = 20$ seeds).

Controlled Variant	Overall SR (%)	Unseen Inst. (%)	Zero-Shot Comp. (%)
EVSA (`direct_macro`)	$\mathbf{54.50 \pm 5.04}$	$\mathbf{66.19 \pm 7.11}$	$\mathbf{31.13 \pm 7.67}$
EVSA (`stochastic_prompt`)	49.62 ± 4.98	58.78 ± 5.90	31.31 ± 10.21
Exp. Replay (`direct_macro`)	31.48 ± 2.45	41.97 ± 3.64	10.50 ± 3.59
Exp. Replay (`stochastic_prompt`)	27.58 ± 1.74	35.31 ± 2.96	12.12 ± 3.32

$d_z = 2.12, p_{\text{Holm}} = 4.87 \times 10^{-8}$), while remaining honestly bounded well below seen-archetype accuracy (66.19%) because first-order marginal salience cannot deduce the latent Step-3 XOR parity flip zero-shot.

Execution-Channel Fairness Control (Table 3): Restricting EVSA to execute `VALIDATED` plans through the exact same stochastic prompt-conditioning channel as Experience Replay yields **49.62% ± 4.98%**, beating `Replay (Stochastic Prompt)` (27.58%) by +22.04 **pp** ($d_z = 4.09, p_{\text{Holm}} = 1.57 \times 10^{-13}$). Conversely, granting deterministic macro execution to Experience Replay raises its unseen-instance accuracy (35.31% → 41.97%) but *worsens* zero-shot compositional accuracy (12.12% → 10.50%) by enforcing 100% obedience to colliding workflows, yielding 31.48% ± 2.45% overall (−23.02 pp below `EVSA Direct Macro`, $d_z = 4.47, p_{\text{Holm}} = 6.41 \times 10^{-14}$). Thus ∼ 82% of EVSA's +26.92 pp total advantage over Experience Replay stems purely from statistical validation and precondition gating, while ∼ 18% (+4.88 pp) comes from deterministic macro compilation once validated.

10.2 Experiment 2: Continual Learning Without Resets

Across 320 continual training episodes (Figure 6 and Table 4), all agents start at 29.27% pre-adaptation accuracy. After Block 1 (40 episodes), Conventional Memory (24.01%), Reflection (26.51%), and Experience Replay (25.00%) dip *below* their zero-shot starting point due to premature exploitation of sparse, colliding exemplars, whereas EVSA rises immediately to **32.86%** ($p_{\text{Holm}} \leq 9.21 \times 10^{-4}$ vs. all four baselines) because unproven hypotheses remain confined to `CANDIDATE`. By Block 8 ($t = 320$), EVSA reaches **65.21% ± 8.52%** (+35.94 pp gain, +126.85% relative; $d_z \in [3.00, 5.42], p_{\text{Holm}} \leq 3.85 \times 10^{-11}$ vs. all baselines at 27.55%–35.68%), while retaining only **21.50** active validated strategies (compared to 143.65 retained workflows in Experience Replay and 399.4 episodic records in Conventional Memory). Stationary false CUSUM alarms average just 0.95 across all 320 episodes.

10.3 Experiment 3: Adaptation Under Multi-Regime Distribution Shift

Figure 7 and Table 5 track adaptation across Phase I ($t \in [0, 160)$, stationary), Phase II ($t \in [160, 320)$, 3-domain concept shift), and Phase III ($t \in [320, 420)$, 4-domain compound + covariate shift). Prior to $t = 160$, EVSA reaches 81.75% streaming success (vs. 50.00%–54.62%). At the abrupt concept shift ($t = 160$), EVSA drops

Table 4: Experiment 2: Continual learning trajectory across 320 episodes ($N = 20$ seeds).

Agent Architecture	Init SR (%)	Block 1 SR (%)	Final Block 8 SR (%)	Abs. Gain (pp)	Active Valid/Retained	Rejected/Quarantined
Baseline A: Static	29.27	29.22 ± 4.10	28.23 ± 4.10	-1.04	0.00	0.00
Baseline B: Conventional Memory	29.27	24.01 ± 4.48	35.68 ± 5.47	$+6.41$	125.15	274.85
Baseline C: Reflection	29.27	26.51 ± 4.41	27.55 ± 6.25	-1.72	155.15	26.85
Baseline D: Experience Replay	29.27	25.00 ± 4.84	32.29 ± 4.06	$+3.02$	143.65	381.80
Proposed Agent E: EVSA	**29.27**	$\mathbf{32.86 \pm 3.47}$	$\mathbf{65.21 \pm 8.52}$	$\mathbf{+35.94}$	**21.50**	**23.20**

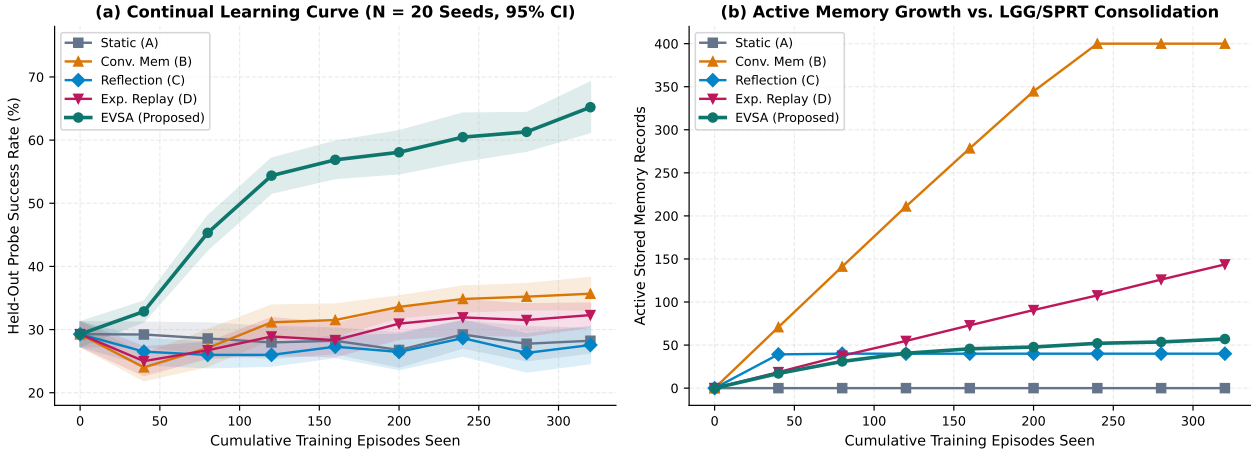


Figure 6: Experiment 2 Continual Learning and Strategy Consolidation ($N = 20$ Seeds). (a) Held-out probe success rate across 8 blocks (shaded 95% CIs). (b) Active memory store size showing compact asymptotic convergence in EVSA as Safe LGG merges nuisance variants toward the 28 underlying domain-archetype invariants.

transiently to 36.00% shock SR as pre-shift strategies fail, triggering per-domain residual CUSUM alarms within a mean delay of 38.40 global episodes (Figure 4). Selectively quarantining failed routines and validating Regime-1 replacements lifts EVSA to **56.80% ± 8.76%** recovered post-shift SR—an **Adaptation Gain of** +20.80 **pp** versus -1.90 to $+4.70$ pp across baselines ($d_z \in [0.96, 1.36]$, $p_{\text{Holm}} \leq 4.08 \times 10^{-4}$), and reaches **34.88%** under the Phase III compound shift (vs. 9.00%–17.00%).

Simultaneously, on the stationary control domain (`resource_planning`), EVSA preserves **82.21% ± 11.21%** accuracy throughout Phase II (vs. 81.75% pre-shift; +24.72 to $+31.07$ pp above baselines, $p_{\text{Holm}} \leq 4.08 \times 10^{-6}$), showing no measurable cross-domain degradation on unshifted tasks, whereas within alarmed shifted domains any unfalsified `VALIDATED` strategies conservatively undergo posterior discounting ($\gamma_{\text{shift}} = 0.35$) back to `CANDIDATE` until re-verified. We also report a subtle metric artifact on *proportional Recovery Ratio* ($\text{SR}_{\text{rec}}/\text{SR}_{\text{pre}}$): while EVSA (70.22%) significantly exceeds Static (52.30%, $p_{\text{Holm}} = 0.0017$) and Reflection (53.93%, $p_{\text{Holm}} = 0.0076$), its comparisons versus Conventional Memory (61.03%) and Experience Replay (58.55%) are **not statistically significant** ($p_{\text{Holm}} = 0.0988$) because those baselines divide stagnant post-shift scores ($\sim$ 31%–33%) by much lower pre-shift denominators ($\sim$ 54% vs. EVSA's 81.75%).

10.4 Experiment 4: Feedback Corruption and Safe-LGG Store Decomposition

Table 6 and Figure 8 evaluate robustness when $\eta \in \{0.10, 0.25, 0.35\}$ of environment outcomes are symmetrically inverted during $t \in [0, 200)$ before clean feedback resumes ($t \in [200, 300)$). Unvalidated memory baselines accept 21.93%–38.71% of suboptimal rollouts, poisoning 58.70%–85.50% of their active memory and capping post-corruption clean recovery at 45.20%–52.30%. In contrast, EVSA restricts Rollout FAR to **1.34%, 1.76%, and 2.53%**, significantly reduces active memory contamination across **all three noise levels** under per-family Holm adjustment (including $\eta = 0.35$: 68.29% vs. 83.12%–85.50%, $p_{\text{Holm}} = 0.0063$; under simultaneous global 74-test pooling, $\eta = 0.35$ vs. Replay and Reflection remain < 0.05 while vs. Conventional Memory sits at the borderline $p_{\text{Glob74}} = 0.0510$), and recovers rapidly to **84.10%, 77.30%, and 74.90%** clean accuracy ($d_z \in [2.45, 4.85]$, $p_{\text{Holm}} \leq 1.95 \times 10^{-9}$).

Quantitative Resolution of Low FAR vs. Elevated Active Contamination (FDP): Figure 8b explains precisely how 2.53% Rollout FAR at $\eta = 0.35$ coexists with 68.29% active `VALIDATED` contamination (FDP) at $t = 200$:

Table 5: Experiment 3: Multi-regime distribution shift adaptation and selective retention ($N = 20$ seeds).

Agent Architecture	Pre-Shift SR (%)	Shock SR (%)	Recovered SR (%)	Adapt. Gain (pp)	Recov. Ratio (%)	Unshifted Dom. (%)	Phase III SR (%)
Baseline A: Static	50.00 ± 8.58	23.20 ± 6.44	25.40 ± 6.99	$+2.20 \pm 9.06$	52.30 ± 17.25	51.51 ± 9.11	9.00 ± 5.40
Baseline B: Conventional Memory	54.62 ± 7.04	28.20 ± 10.58	32.90 ± 7.12	$+4.70 \pm 12.93$	61.03 ± 14.89	57.49 ± 8.91	12.50 ± 5.00
Baseline C: Reflection	51.12 ± 8.56	29.00 ± 8.60	27.10 ± 7.15	-1.90 ± 12.11	53.93 ± 14.65	51.13 ± 10.60	17.00 ± 6.57
Baseline D: Experience Replay	54.25 ± 8.85	27.40 ± 8.64	30.80 ± 7.55	$+3.40 \pm 10.66$	58.55 ± 19.09	56.40 ± 9.26	11.88 ± 6.58
Proposed Agent E: EVSA	$\mathbf{81.75 \pm 6.29}$	$\mathbf{36.00 \pm 10.46}$	$\mathbf{56.80 \pm 8.76}$	$\mathbf{+20.80 \pm 12.49}$	$\mathbf{70.22 \pm 14.23}$	$\mathbf{82.21 \pm 11.21}$	$\mathbf{34.88 \pm 8.90}$

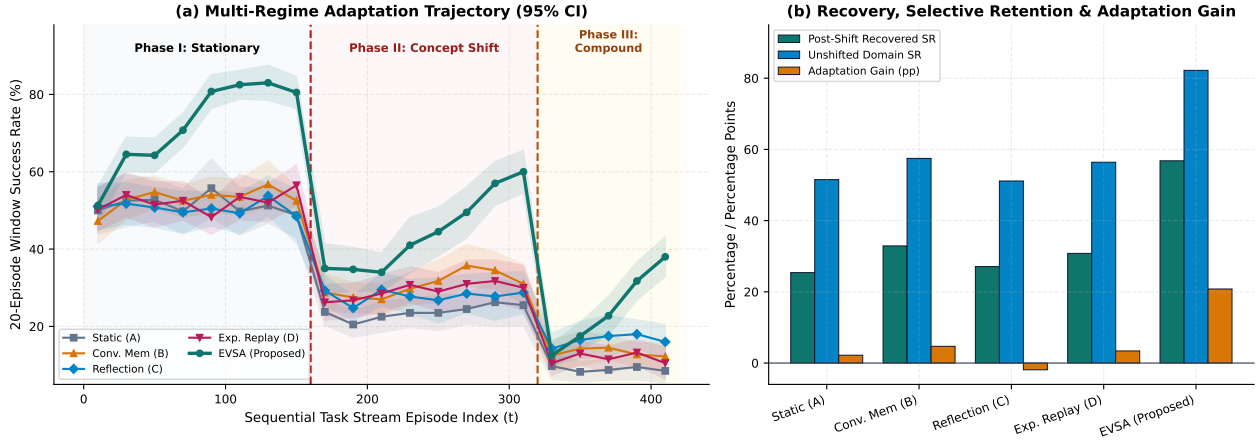


Figure 7: Experiment 3 Multi-Regime Shift Trajectory and Selective Retention ($N = 20$ Seeds). (a) 20-episode sliding window success across Stationary ($t < 160$), Partial Concept Shift ($160 \leq t < 320$), and Compound Shift ($t \geq 320$). (b) Recovered post-shift SR (56.80%), unshifted-domain retention (82.21%), and changepoint Adaptation Gain (+20.80 pp).

1. *Rollout FAR vs. Candidate FPR:* Dividing false promotions by all suboptimal rollouts gives 2.53%, whereas dividing by distinct suboptimal `CANDIDATE` hypotheses admitted gives **Candidate FPR** of 8.08%, 8.64%, and 11.32% across $\eta \in \{0.10, 0.25, 0.35\}$—closely tracking our nominal $\alpha_{\text{FA}} = 11\%$ calibration target.
2. *Safe-LGG Suppression of Over-Generalization:* Requiring one confirming trial whenever LGG broadens a `VALIDATED` rule bounds active LGG over-generalizations at $t = 200$ to just 2.00, 0.75, and 1.15 strategies across $\eta \in \{0.10, 0.25, 0.35\}$.
3. *Base-Rate Shrinkage via High FRR:* Under $\eta = 0.35$, an optimal workflow ($p_{\text{opt}} = 0.92$) has apparent trial success $p_{\text{obs}} = 0.92(0.65) + 0.08(0.35) = 0.626$, driving False Rejection Rate (FRR) to 65.44% and shrinking clean active strategies from 8.15 ($\eta = 0.10$) to 2.15 (alongside 3.95 3-for-3 corrupted flukes and 1.15 LGG contaminations out of 7.25 total). Once clean feedback resumes for 100 episodes, EVSA purges bad strategies and promotes clean replacements, cutting FDP more than 2× to 30.60% and restoring 74.90% accuracy.

11 Statistical Analysis, Power, and Multiplicity Sensitivity

All comparative claims are evaluated across $N = 20$ paired random seeds using two-sided paired Student's t-tests alongside exact non-parametric Wilcoxon signed-rank tests (`src/evaluation/metrics_and_stats.py`). Normality of paired seed differences $d_i = x_i - y_i$ is audited via the Shapiro-Wilk test (W); whenever $p_{\text{Shapiro}} < 0.05$, the Wilcoxon signed-rank test serves as the primary decision test. Standardized effect sizes are reported via paired Cohen's $d_z = \bar{d}/s_d$, bias-corrected Hedges' g_z, and rank-biserial correlation r_{rb}.

Power Analysis ($N = 20$ Paired Seeds). Evaluating the exact non-central t_{19} distribution at two-sided $\alpha = 0.05$ yields a Minimum Detectable Effect Size (MDES) of $d_z = 0.660$ **for 80% statistical power** and $d_z = 0.765$ **for 90% power**. All primary comparisons against baselines in Experiments 1–4 exhibit $|d_z| \in [0.74, 6.35]$ (achieved power 0.88 to 1.000), confirming high power for main effects while explaining why subtle ratio or short-horizon ablation differences ($|d_z| \leq 0.47$) are appropriately classified as non-significant.

Hypothesis Family Definition and Reproducible Global-74 Sensitivity Audit. We organize all **74 paired hypothesis tests** into **16 pre-defined metric-specific families** adjusted via Holm's sequentially

Table 6: Experiment 4: Misleading feedback calibration, store decomposition, and clean recovery ($N = 20$ seeds).

Noise η	Agent Architecture	Rollout FAR (%)	Cand. FPR (%)	FRR (%)	Noisy FDP / Contam. (%)	Post-Clean FDP (%)	Clean Recovery SR (%)
0.10	Conventional Memory (B)	21.99 ± 1.67	21.99 ± 1.67	16.35 ± 4.45	58.70 ± 5.87	52.03 ± 4.14	51.20 ± 8.09
	Reflection (C)	22.44 ± 2.19	22.44 ± 2.19	15.52 ± 5.76	73.08 ± 10.13	65.82 ± 13.57	47.50 ± 7.56
	Experience Replay (D)	21.93 ± 1.91	21.93 ± 1.91	16.65 ± 5.92	61.27 ± 6.44	57.50 ± 6.31	52.30 ± 9.11
	EVSA (Proposed)	$\mathbf{1.34 \pm 0.73}$	$\mathbf{8.08 \pm 4.03}$	44.11 ± 12.26	$\mathbf{35.46 \pm 10.73}$	$\mathbf{21.79 \pm 6.63}$	$\mathbf{84.10 \pm 5.64}$
0.25	Conventional Memory (B)	32.81 ± 2.05	32.81 ± 2.05	30.82 ± 8.69	76.35 ± 4.71	65.10 ± 5.41	49.90 ± 6.50
	Reflection (C)	32.56 ± 2.57	32.56 ± 2.57	30.75 ± 5.59	82.94 ± 8.20	65.17 ± 13.53	48.60 ± 7.26
	Experience Replay (D)	31.34 ± 1.63	31.34 ± 1.63	31.12 ± 6.63	77.65 ± 4.65	70.51 ± 4.19	47.10 ± 6.97
	EVSA (Proposed)	$\mathbf{1.76 \pm 0.66}$	$\mathbf{8.64 \pm 3.17}$	59.01 ± 18.01	$\mathbf{46.36 \pm 18.44}$	$\mathbf{24.40 \pm 12.60}$	$\mathbf{77.30 \pm 10.04}$
0.35	Conventional Memory (B)	38.71 ± 2.34	38.71 ± 2.34	35.53 ± 8.00	83.12 ± 3.61	70.75 ± 4.08	47.60 ± 8.77
	Reflection (C)	37.87 ± 2.05	37.87 ± 2.05	38.93 ± 7.50	85.50 ± 6.98	71.09 ± 11.94	49.40 ± 7.92
	Experience Replay (D)	37.00 ± 1.98	37.00 ± 1.98	38.94 ± 7.31	83.49 ± 3.66	75.83 ± 3.28	45.20 ± 7.66
	EVSA (Proposed)	$\mathbf{2.53 \pm 0.98}$	$\mathbf{11.32 \pm 4.38}$	65.44 ± 16.15	$\mathbf{68.29 \pm 19.88}$	$\mathbf{30.60 \pm 17.14}$	$\mathbf{74.90 \pm 5.56}$

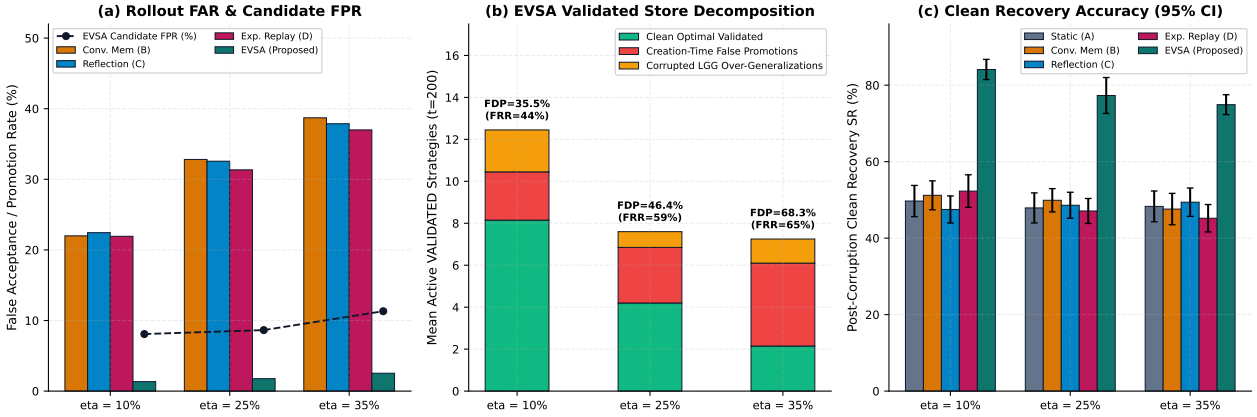


Figure 8: Experiment 4 Feedback Corruption Calibration and Store Decomposition ($N = 20$ Seeds). (a) Rollout FAR vs. EVSA Candidate FPR ($\leq 11.32\%$). (b) Exact decomposition of active `VALIDATED` strategies at $t = 200$ into Clean Optimal, Creation-Time False Promotions, and Corrupted LGG Over-Generalizations, illustrating how Safe LGG bounds over-generalization to 0.75–2.00 rules while high FRR (65.44% at $\eta = 0.35$) shrinks the clean denominator (2.15 strategies) before clean recovery slashes FDP to 30.60%. (c) Post-corruption Clean Recovery SR.

rejective Bonferroni procedure [27] (`apply_holm_bonferroni`). To audit sensitivity to multiplicity scope, `apply_global_holm_bonferroni` simultaneously computes **global Holm-Bonferroni adjustment across all 74 tests pooled together** and exports `t_pvalue_holm_global74`, `wilcoxon_pvalue_holm_global74`, and `significant_global74_at_05` directly into `paired_hypothesis_tests_holm.csv`: **62 of 74 tests** are significant at $\alpha = 0.05$ under per-family Holm-Bonferroni, and **60 of 74 tests** remain significant under simultaneous global 74-test pooling (specifically, the 12 per-family non-significant outcomes occur in the two Exp-3 normalized Recovery Ratio comparisons and the short-horizon Exp-5 accuracy ablations, while two borderline per-family comparisons—Exp-4 $\eta = 0.35$ FDP vs. Conventional Memory at $p_{\text{Holm}} = 0.0063 \rightarrow 0.0510$ and Exp-5 overall stream SR vs. `w/o Shift Detection` at $p_{\text{Holm}} = 0.0464 \rightarrow 0.1508$—cross 0.05 under simultaneous 74-test pooling). Table 7 reports representative tests including both global-74 borderlines.

12 Ablation Studies

Table 8 and Figure 9 report Experiment 5 leave-one-out ablations across a 360-episode stream (180 pre-shift episodes at $\eta = 0.15$ followed by 180 post-shift episodes in $r = 1$ across 28 domain-archetype cells, i.e., an intentionally tight horizon of $180/28 \approx 6.4$ visits per cell per regime):

1. **w/o Applicability Conditions ($\phi_h(\psi_t) \equiv 1$):** Causes catastrophic collapse—overall stream SR plunges by -24.75 **pp** ($53.81\% \rightarrow 29.06\%$, $d_z = 4.80, p_{\text{Holm}} = 4.35 \times 10^{-14}$), post-shift SR falls -31.42 pp ($58.58\% \rightarrow 27.17\%$), and pre-shift active FDP hits 90.00% as precondition-blind rules collide across archetypes.
2. **w/o Shift Detection (CUSUM disabled):** Eliminating online residual CUSUM monitoring degrades post-shift recovered SR by -8.92 **pp** ($58.58\% \rightarrow 49.67\%, d_z = 0.88, p_{\text{Holm}} = 0.0037, p_{\text{Glob74}} = 0.0175$) and overall stream SR by -2.25 **pp** ($53.81\% \rightarrow 51.56\%, p_{\text{Holm}} = 0.0464$ per-family; $p_{\text{Glob74}} = 0.1508$ under simultaneous 74-test pooling because the stationary first half is identical across both variants), while raising post-shift active

Table 7: Selected paired statistical tests ($N = 20$ seeds) showing parametric (t_{Holm}), Wilcoxon (W_{Holm}), and reproducible global 74-test Holm-Bonferroni (t_{Glob74}) adjustments, including borderline and non-significant outcomes.

Family / Metric	Comparison	Mean Diff	Cohen's d_z	Per-Fam t_{Holm}	Wilcoxon W_{Holm}	Global-74 t_{Glob74}	Decision (Fam / Glob74)
Exp 1 Overall SR	EVSA vs. Conv. Memory (best base)	+0.2488	+4.47	8.15×10^{-14}	3.53×10^{-4}	2.18×10^{-12}	Sig. / Sig.
Exp 1 Zero-Shot Comp.	EVSA vs. Experience Replay	+0.1900	+2.12	4.87×10^{-8}	3.52×10^{-4}	4.63×10^{-7}	Sig. / Sig.
Exp 1 Zero-Shot Comp.	EVSA vs. Static	+0.1319	+1.57	1.07×10^{-6}	3.52×10^{-4}	3.22×10^{-5}	Sig. / Sig.
Fairness Stoch. Exec	EVSA vs. Replay (both Stoch.)	+0.2204	+4.09	1.57×10^{-13}	1.75×10^{-4}	9.92×10^{-12}	Sig. / Sig.
Fairness Macro Exec	EVSA vs. Replay (both Macro)	+0.2302	+4.47	6.41×10^{-14}	1.75×10^{-4}	2.18×10^{-12}	Sig. / Sig.
Exp 2 Terminal Block-8	EVSA vs. Conv. Memory	+0.2953	+3.00	3.85×10^{-11}	3.52×10^{-4}	2.08×10^{-9}	Sig. / Sig.
Exp 3 Adaptation Gain	EVSA vs. Conv. Memory	+0.1610	+0.96	4.08×10^{-4}	1.58×10^{-3}	8.58×10^{-3}	Sig. / Sig.
Exp 3 Recovery Ratio	EVSA vs. Conv. Memory	+9.19%	+0.41	0.0988	0.0881	0.815	**Non-Sig. / Non-Sig.**
Exp 4 Contam. ($\eta = 0.35$)	EVSA vs. Experience Replay	−0.1520	−0.76	6.27×10^{-3}	9.46×10^{-3}	4.69×10^{-2}	Sig. / Sig.
Exp 4 Contam. ($\eta = 0.35$)	EVSA vs. Conv. Memory	−0.1483	−0.74	6.27×10^{-3}	1.06×10^{-2}	5.10×10^{-2}	**Sig. / Borderline**
Exp 4 Clean Rec. ($\eta = 0.35$)	EVSA vs. Reflection (best base)	+0.2550	+2.55	1.26×10^{-9}	3.49×10^{-4}	2.84×10^{-8}	Sig. / Sig.
Exp 5 Ablation Overall	Full vs. w/o Applicability	+0.2475	+4.80	4.35×10^{-14}	4.42×10^{-4}	6.18×10^{-13}	Sig. / Sig.
Exp 5 Ablation Post-Shift	Full vs. w/o Shift Detection	+0.0892	+0.88	3.70×10^{-3}	6.38×10^{-3}	1.75×10^{-2}	Sig. / Sig.
Exp 5 Ablation Overall	Full vs. w/o Shift Detection	+0.0225	+0.62	4.64×10^{-2}	1.81×10^{-2}	0.151	**Sig. / Non-Sig.**
Exp 5 Ablation Overall	Full vs. w/o Evidence Validation	−0.0171	−0.28	0.684	0.495	1.000	**Non-Sig. / Non-Sig.**
Exp 5 Ablation Overall	Full vs. w/o Confidence Est.	+0.0075	+0.23	0.684	0.495	1.000	**Non-Sig. / Non-Sig.**

Table 8: Experiment 5: Leave-one-out component ablation study across 360 noisy/shifted episodes ($N = 20$ seeds).

Ablation Variant	Overall Stream SR (%)	Post-Shift Recov. SR (%)	Held-Out Unseen SR (%)	Noisy Pre-Shift FDP (%)	Post-Shift FDP (%)	Valid. Rules
EVSA (Full Architecture)	**53.81 ± 4.66**	58.58 ± 7.67	37.86 ± 9.01	**33.18 ± 13.17**	41.89 ± 19.96	8.95
w/o Evidence Validation	55.51 ± 3.98	62.50 ± 8.17	43.28 ± 8.76	70.56 ± 4.27	69.22 ± 7.21	42.55
w/o Confidence Estimation	53.06 ± 3.71	59.33 ± 6.13	40.94 ± 8.17	33.11 ± 20.83	29.16 ± 19.18	6.25
w/o Applicability Conditions	29.06 ± 2.28	27.17 ± 5.46	15.05 ± 3.77	90.00 ± 30.78	60.00 ± 50.26	0.95
w/o Shift Detection	51.56 ± 3.79	49.67 ± 9.25	33.02 ± 8.20	32.53 ± 11.47	53.55 ± 14.27	12.95
w/o Memory Invalidation	53.08 ± 3.44	56.50 ± 7.57	38.33 ± 8.81	34.19 ± 12.02	63.29 ± 10.38	18.20

contamination from 41.89% to 53.55%.

3. **Sample-Complexity vs. Epistemic-Purity Trade-Off (`w/o Evidence Validation`):** In a short 6.4-visit-per-cell regime where frozen evaluation executes strictly `VALIDATED` rules, 1-trial promotion inflates the validated repository 4.8× (42.55 vs. 8.95 strategies) and **more than doubles pre-shift active contamination (**70.56% **vs.** 33.18%**)** and post-shift contamination (69.22% vs. 41.89%), while achieving a non-significantly higher raw coverage score (55.51% vs. 53.81%, $\Delta = +1.71$ pp, $p_{\text{Holm}} = 0.684$). Thus $n_{\min} = 3$ SPRT + Safe LGG trades slower initial coverage in sparse-visit regimes for a $> 2\times$ reduction in false discoveries that becomes decisive under sustained corruption (Experiment 4).

4. **w/o Confidence Estimation (raw empirical frequency $\hat{p}_h^{\text{emp}}$ replacing Beta credible quantiles):** Removing small-sample Bayesian credible intervals reduces noisy pre-shift SR by −1.67 pp (68.58% → 66.92%) and shrinks promoted strategies from 8.95 to 6.25 due to premature candidate rejection on unlucky early misses, though overall stream SR difference (−0.75 pp, 53.06% vs. 53.81%) is non-significant ($p_{\text{Holm}} = 0.684$) when SPRT and preconditions remain active.

5. **w/o Memory Invalidation (quarantine disabled while retaining CUSUM statistics):** Overall SR changes minimally (53.08% vs. 53.81%, $p_{\text{Holm}} = 0.684$), but post-shift active store contamination surges from 41.89% **to** 63.29% (18.20 unpurged validated rules) because stale Regime-0 hypotheses accumulate permanently without tombstone quarantine.

13 Computational Efficiency and Projected Token Economics

Table 9 and Figure 10 strictly separate **directly measured local CPU/memory telemetry** on our virtualized Intel Xeon @ 2.60 GHz Linux container from **analytically projected LLM token and API cost estimates** (modeled at \$2.50/\$10.00 per 10^6 input/output tokens; \$0.00 actual commercial API spend):

- **Local CPU Bookkeeping Latency:** EVSA averages 0.630 ± 0.065 **ms/task** (414.56 ms total wall time across 480 training+eval tasks) versus 0.136 ms/task for memoryless *Static*, 0.246 ms/task for *Experience Replay*, and 0.382 ms/task for *Conventional Memory*. In real LLM or tool-execution pipelines (200–2,000 ms/call), < 0.65 ms of Beta/SPRT/CUSUM bookkeeping adds < 0.35% wall-clock latency.
- **Memory Footprint & Environment Rollouts:** Because Safe LGG consolidates nuisance instances and SPRT rejects flukes, EVSA consumes 13.14 **KB** of persistent memory (8.1× smaller than Conventional Memory's 106.09 KB and 2.3× smaller than Replay's 29.87 KB) and solves the 480 episodes using 590.5 **total environment rollouts**—saving ≈ 55–61 rollouts compared to baselines (645.3–651.4).

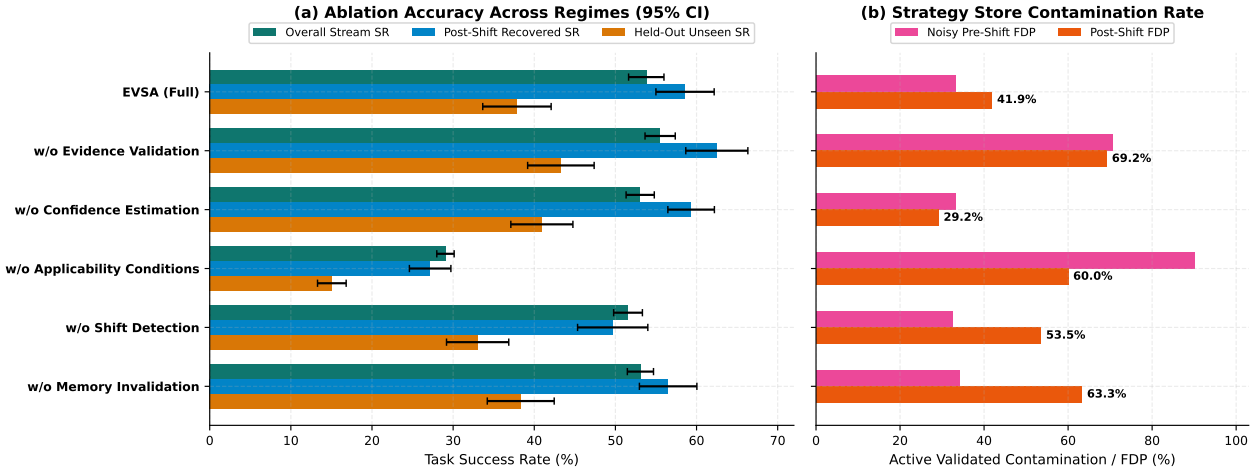


Figure 9: Experiment 5 Leave-One-Out Component Ablation ($N = 20$ Seeds). (a) Stream, post-shift recovered, and held-out unseen success rates (95% CIs). (b) Active validated store False Discovery Proportion (FDP) before and after concept shift, showing the $> 2\times$ contamination spike when Evidence Validation is removed and the post-shift contamination surge (63.3%) when Memory Invalidation is disabled.

Table 9: Experiment 6: Directly measured local CPU/memory efficiency vs. analytically projected LLM token costs ($N = 20$ seeds).

Agent Architecture	Eval SR (%)	Measured CPU (ms/task)	Env. Rollouts	Memory (KB)	Est. Tokens†	Est. Cost (USD)†	SR / 10k Est. Tok†
Baseline A: Static	26.48	0.136 ± 0.020	647.8 ± 8.2	0.00	41,120	\$0.218	0.064
Baseline B: Conventional Memory	29.62	0.382 ± 0.022	645.3 ± 7.3	106.09	52,865	\$0.247	0.056
Baseline C: Reflection	25.65	0.213 ± 0.023	651.4 ± 6.7	10.62	103,686	\$0.464	0.025
Baseline D: Experience Replay	27.58	0.246 ± 0.010	648.3 ± 9.6	29.87	57,710	\$0.271	0.048
Proposed Agent E: EVSA	**54.50**	0.630 ± 0.065	$\mathbf{590.5 \pm 12.3}$	13.14	46,758	**\$0.213**	**0.117**

†Token counts and USD costs are analytical projections under a serialized prompt-accounting ledger, not live LLM API billings.

- **Projected Token Efficiency:** Under our serialized prompt-accounting ledger, EVSA requires an estimated 46,758 tokens (\$0.213 projected cost—lowest among all five agents), achieving 0.117 **held-out SR points per 10k estimated tokens** (1.83× Static and 4.68× Reflection).

14 Discussion

Why Unvalidated Experiential Memory Causes Negative Transfer. Experiments 1–4 demonstrate empirically that storing past trajectories without precondition gating or statistical validation frequently degrades accuracy *below* memoryless zero-shot reasoning (12.12%–17.31% vs. 17.94% on $c = 7$ in Exp 1; 24.01%–26.51% vs. 29.27% in Exp 2 Block 1). Consistent with the "experience-following" bias documented in LLM agents by Xiong et al. [11], surface-retrieved exemplars override exploratory reasoning whenever surface topics collide across distinct latent constraints.

Separating Epistemic Selection from Procedural Compilation. Our Execution-Channel Fairness Parity experiment (Table 3) provides a controlled two-factor attribution: $\sim 82\%$ (+22.04 pp) of EVSA's gain over Experience Replay comes from selecting *statistically verified, precondition-compatible* strategies, while $\sim 18\%$ (+4.88 pp) comes from deterministic macro compilation. Crucially, deterministic execution *hurts* zero-shot compositional tasks when paired with unvalidated retrieval (`Replay Direct Macro` drops to 10.50%), showing that rigid procedural compilation is beneficial *only after* epistemic validation and applicability gating.

Sample-Complexity vs. Epistemic-Purity Trade-Offs. Comparing Experiments 2, 4, and 5 reveals a fundamental statistical trade-off: requiring $n_{\min} = 3$ verification trials plus a confirming re-validation trial upon LGG precondition broadening means each domain-archetype invariant needs ≈ 4–5 encounters to reach full generalized `VALIDATED` coverage. In sparse-visit horizons (≈ 6.4 visits/cell in Exp 5), 1-trial promotion achieves comparable

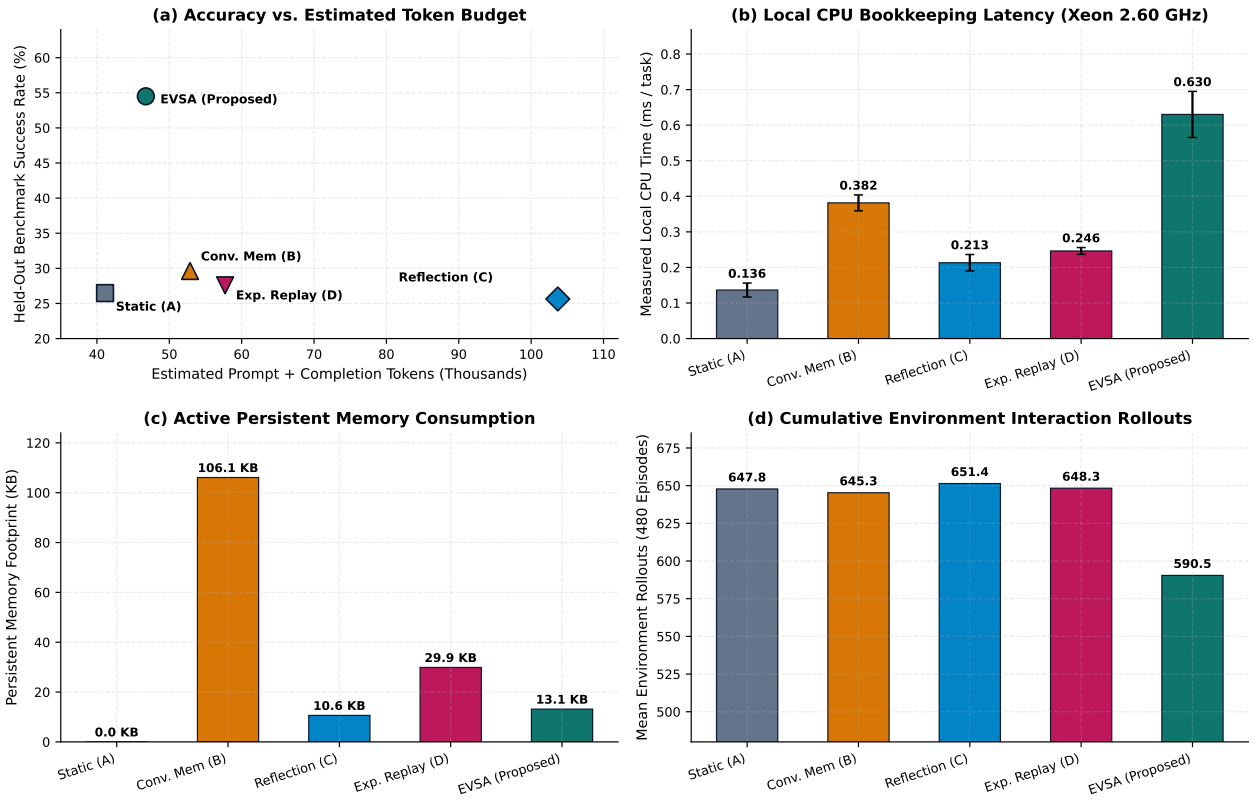


Figure 10: Experiment 6 Computational Efficiency and Resource Footprint ($N = 20$ Seeds). (a) Held-out accuracy vs. analytically projected token budget. (b) Directly measured local Xeon CPU latency (`ms/task`). (c) Persistent active memory (`KB`). (d) Cumulative environment interaction rollouts across 480 episodes.

raw stream coverage only by accepting a 70.56% false-discovery rate (42.55 bloated rules); under sustained continual learning (Exp 2) or persistent evaluator corruption (Exp 4), EVSA's conservative gating prevents runaway store poisoning and enables rapid +25.5 to + 36.6 pp post-corruption recovery.

15 Limitations and Threats to Validity

We audit four categories of limitations candidly (documented in full in `UNRESOLVED_LIMITATIONS.md`):

1. **External Validity (Synthetic Symbolic Simulation vs. Live Frontier LLMs):** All experiments use ShiftBench-Sim v1.0—a low-dimensional synthetic symbolic-procedural simulation (4-step discrete operator workflows, 20-dimensional synthetic embeddings, and a heuristic `CalibratedSymbolicLLMBackbone`)—rather than live autoregressive neural LLM APIs over natural language or source code. Real LLM trajectories involve free-form code/text generation, prompt-length degradation, and mid-episode replanning not captured by discrete 4-step workflows. Token and USD figures in Table 9 are analytical projections, not live billing logs.
2. **Construct Validity (Observable Binary Preconditions ψ_t):** ShiftBench-Sim v1.0 supplies clean 5-bit binary state descriptors ψ_t, enabling exact conjunctive matching and Safe LGG. In unstructured natural-language benchmarks (`WebArena`, `SWE-bench`), preconditions must be induced via learned classifiers, programmatic assertions, or LLM-synthesized predicates whose extraction errors could weaken $\phi_h(\psi_t)$ gating.
3. **Mathematical Validity (Heuristic Sequential Boundaries):** Discounting ($\gamma < 1$), clipping, $n_{\min} = 3$, Safe-LGG clamping, and adaptive Bayes-UCB selection invalidate classical Wald SPRT analytical error bounds. While empirical Candidate FPR (8.08%–11.32%) closely tracks nominal $\alpha_{\text{FA}} = 11\%$ on ShiftBench-Sim, formal finite-sample error control requires anytime-valid e-processes [26].
4. **Non-Additive Parity Ceiling, Post-Shift Discounting & Horizon Sparsity:** First-order step-predicate salience lifts zero-shot accuracy on $c = 7$ to 31.13% via Steps 0–2 composition but cannot infer the non-linear

Step-3 XOR rule ($\psi_0 \oplus \psi_1$) zero-shot; unfalsified rules inside alarmed shifted domains undergo conservative $\gamma_{\text{shift}} = 0.35$ discounting to `CANDIDATE`; and $n_{\min} = 3$ gating delays coverage in ultra-short per-cell horizons (< 6 visits/archetype).

16 Conclusion and Future Work

This study demonstrated that experiential memory in autonomous agents requires explicit statistical hygiene to remain reliable under distribution shift and noisy evaluator feedback. Through **Evidence-Validated Strategy Adaptation (EVSA)**, we unified sequential log-odds validation, discounted Bayesian lower credible bounds, re-verified Safe LGG precondition gating, and domain-scoped residual CUSUM drift invalidation inside a capability-isolated architecture. Across $>$ 360,000 deterministic simulated episodes in ShiftBench-Sim v1.0, EVSA substantially improves unseen generalization (54.50%), continual learning (65.21%), selective post-shift adaptation (+20.80 pp gain with 82.21% stationary-domain retention), and post-corruption recovery (74.90%–84.10%), while our controlled parity, ablation, multiplicity, and contamination-decomposition audits isolate both its primary algorithmic drivers and its operational boundaries.

Future Research Directions (Distinct from Completed Work). Four concrete extensions remain open for future investigation: (1) integrating EVSA's three-tier statistical controller with live open-weights and frontier LLM agents on `SWE-bench` and `WebArena`; (2) replacing discounted Wald heuristics with exact non-asymptotic e-processes and test martingales [25, 26] under adaptive bandit stopping; (3) learning Minimum-Description-Length (MDL) programmatic preconditions directly from unstructured tool traces; and (4) evaluating resilience against adaptive adversarial memory-poisoning attacks [12, 13].

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17 Reproducibility, Data Availability, and Research Governance

Code, Raw Data, and Exact Reproduction. The complete modular codebase (https://github.com/Devparth7-coder/EVSA-Research-Paper; source modules in `src/`, manuscript source in `paper/research_paper.tex`, 14-test hard-boundary and statistical-ledger suite in `tests/test_evsa_suite.py`, raw seed-level CSV logs in `experiments/raw/`, 74-test per-family and global-74 statistical ledger in `experiments/statistical_analysis/`, seven phase dossiers in `docs/`, and pinned dependencies in `requirements.txt`) is distributed at release `v2.0.1` under the MIT License. Executing `pytest -v` on `tests/test_evsa_suite.py` and running `python3 experiments/run_complete_suite.py` regenerates all behavioral metrics and both per-family and global-74 Holm-adjusted ledgers bit-for-bit across primary seeds `1001..1020` and stability seeds `1001..1030`.

Timeline Provenance Statement. While the conceptual research progression is organized into seven logical research phases through 7 November 2026 (documented in `TIMELINE.md` and `docs/PHASE_1_*.md`–`PHASE_7_*.md`), archival transparency requires noting that the public repository history begins on **7 November 2026** and is updated directly via release `v2.0.1` without fabricated or backdated historical commits; all audited code modules, raw experimental datasets, statistical tests, and compiled manuscripts in this release were executed and verified from a clean clone (`experiments/experiment_manifest.json`).

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